

**SMARTFLOW: AN AI-DRIVEN AGENT-BASED SIMULATION OF
AUTONOMOUS VEHICLES AND PEDESTRIANS WITH REINFORCEMENT
LEARNING–BASED TRAFFIC SIGNAL OPTIMIZATION**



**A Capstone Project Presented to
Faculty of the Institute of Computing
Davao del Norte State College
New Visayas, Panabo City**

A Capstone Project Presented by:

**Johnfel Anthony B. Caredo
Prille Vincent Salibay
Karen B. Solano**

June 2026

CHAPTER I

INTRODUCTION

Background of the Study

The rapid urbanization and increasing population density in cities worldwide have created significant challenges in traffic management, including traffic congestion, environmental pollution, increased fuel consumption, and inefficiencies in urban mobility services [1]. As the number of vehicles continues to rise, transportation infrastructures in many urban areas struggle to accommodate growing traffic demand, resulting in longer travel times and substantial economic losses. These challenges have made traffic congestion one of the most pressing transportation issues faced by cities worldwide, highlighting the need for more effective and adaptive traffic management strategies [2].

Despite the implementation of various traffic management approaches, many urban centers continue to rely on conventional traffic signal control systems that operate using fixed or pre-timed schedules [3]. These systems are often unable to adapt to changing traffic conditions, causing inefficient traffic flow and increased congestion [4]. As a result, green signal phases may be allocated to underutilized lanes while heavily congested approaches experience prolonged delays. This lack of adaptability not only reduces traffic throughput but also contributes to economic losses, commuter dissatisfaction, and increased greenhouse gas emissions. Consequently, transportation researchers and traffic management agencies are exploring more intelligent and proactive approaches capable of responding to dynamic traffic environments.

To address these global traffic management challenges, Artificial Intelligence (AI) has emerged as a promising technology for developing intelligent transportation systems and adaptive traffic control solutions [2]. Recent

studies have demonstrated that AI-driven signal control systems, particularly those utilizing Deep Reinforcement Learning (DRL), significantly outperform traditional fixed-time controllers by dynamically adjusting signal timings according to real-time traffic conditions, reducing average traffic delays by up to 30% [5]. Furthermore, the scientific community has extensively explored the integration of Reinforcement Learning (RL) and multi-agent systems to solve complex traffic dilemmas. Research has shown that DRL applied within the Simulation of Urban Mobility (SUMO) environment can reduce intersection waiting times by as much as 33% compared to conventional control systems [6]. To improve the realism and adaptability of intelligent traffic control models, microscopic traffic simulation frameworks have been developed to enable AI agents to generalize across dynamic and previously unseen traffic scenarios [7]. Additionally, benchmarking studies have reported that RL-enhanced traffic management approaches can reduce delays by up to 80% in high-demand traffic situations while contributing to environmental sustainability through reduced vehicle idle times and lower greenhouse gas emissions [8]. Collectively, these developments demonstrate the growing role of AI-driven simulation and Reinforcement Learning as effective solutions for modern traffic management and urban mobility challenges.

The traffic crisis in the Philippines has reached a critical threshold, with recent data from 2025 identifying the country as one of the most congested in Asia [8]. Projections from the Japan International Cooperation Agency (JICA) indicate that without significant technological intervention, the volume of vehicles in major Philippine urban centers will increase by more than 2.25 times by 2035, further exacerbating the billions of pesos lost daily in economic productivity. To combat this, the Philippine Development Plan (PDP) 2023–2028 emphasizes the "digital transformation" of physical connectivity, calling for the adoption of smart transport infrastructure to improve urban mobility and climate resilience [9]. However, most current local implementations remain limited to surveillance and manual enforcement rather than autonomous optimization. This national gap highlights the urgency for locally-developed, AI-driven solutions like

SMARTFLOW that can adapt to the unique road constraints and behavioral patterns of Filipino commuters.

The study is focused on a primary high-volume intersection and its immediately adjacent road networks in Tagum City, Davao del Norte. This specific locale was selected because it represents a critical "bottleneck" where diverse road constraints—such as public utility vehicle (PUV) stops, pedestrian crossings, and narrow lane widths—frequently disrupt traffic flow. To maintain high fidelity, the simulation environment is modeled as a real-world-inspired simulation model of the actual site, using field observations and visual references to replicate real-world geometry and agent behaviors. By limiting the scope to a single intersection and its connected road segments, the study ensures a deep, granular analysis of traffic impact and signal efficiency without the computational overhead of a city-wide model.

Despite the implementation of modern traffic management strategies, urban centers like Tagum City continue to struggle with the unpredictable impacts of road constraints. Currently, traffic management offices often operate reactively responding to congestion only after it has already formed due to incidents such as road construction, lane closures, or accidents. There is a significant lack of proactive decision-support tools that can simulate and analyze how specific physical constraints may influence congestion buildup and traffic flow. Without the ability to "test" road closures or evaluate signal effectiveness in a realistic simulation environment before actual implementation, authorities are forced to rely on trial-and-error, which leads to avoidable economic losses and commuter dissatisfaction. This study addresses the need for an AI-driven, agent-based platform that can analyze the ripple effects of road constraints and provide optimized signal responses using Reinforcement Learning to mitigate bottlenecks before they manifest in the real world.

This study proposes SMARTFLOW, an AI-driven simulation-based decision-support platform designed to analyze the impact of road constraints and

optimize traffic flow through Reinforcement Learning (RL). Utilizing the Simulation of Urban Mobility (SUMO) and the TraCI API, the system implements an agent-based model where vehicles and pedestrians act as autonomous entities with intelligent behaviors. Unlike traditional static traffic management systems, SMARTFLOW enables traffic managers to simulate various traffic scenarios, such as road construction, accidents, lane closures, and flooding, to proactively evaluate congestion patterns and traffic signal responsiveness. The RL component dynamically adjusts traffic signal timings based on current traffic conditions, providing a data-driven approach to traffic management and urban mobility planning.

This study aligns with Sustainable Development Goal (SDG) 9, which promotes industry, innovation, and resilient infrastructure, and SDG 11, which focuses on creating sustainable cities and communities through improved urban mobility and transportation systems. Furthermore, the study supports the Davao del Norte State College (DNSC) Research, Development, and Extension (RDE) Agenda on Information and Communication Technology for Development by applying Artificial Intelligence, simulation technologies, and intelligent decision-support systems to address transportation and infrastructure challenges.

Generally, SMARTFLOW functions as a simulation-based traffic analysis and decision-support system that enables users to model traffic scenarios and evaluate the effects of different road constraints on traffic flow. The system utilizes Reinforcement Learning to generate adaptive traffic signal decisions, allowing traffic managers and researchers to assess traffic performance, compare signal control strategies, and identify potential congestion mitigation measures before real-world implementation.

Objectives of the Study

General Objective

To develop SMARTFLOW, an AI-driven agent-based traffic simulation and decision-support system that models autonomous vehicle and pedestrian behavior and utilizes Reinforcement Learning for adaptive traffic signal optimization under varying traffic and road constraint conditions.

Specific Objectives

1. Design a traffic simulation environment based on a selected intersection in Tagum City, incorporating road layout, traffic flow patterns, and pedestrian behavior.
2. Develop an agent-based model where vehicles and pedestrians are represented as autonomous entities with heterogeneous behaviors and dynamic interaction within the simulation environment.
3. Implement dynamic traffic generation using scenario-based and configurable traffic demand models instead of relying on static predefined inputs.
4. Develop a Reinforcement Learning–based traffic signal control component that optimizes signal timing based on real-time traffic conditions.
5. Simulate and evaluate multiple traffic and road constraint scenarios, including varying traffic densities, pedestrian demand levels, emergency vehicle conditions, lane closures, road construction, and accidents, to assess the adaptability and performance of the reinforcement learning–based traffic signal control system.

Significance of the Study

This study contributes to the advancement of intelligent transportation systems by integrating agent-based modeling and reinforcement learning into a localized traffic simulation and decision-support platform. The proposed system enables dynamic interaction between autonomous vehicle and pedestrian agents while

allowing adaptive traffic signal optimization through Reinforcement Learning (RL). Unlike traditional traffic management approaches that rely on static signal timing and reactive congestion handling, the system provides a simulation-based environment where various traffic and road constraint scenarios can be analyzed before real-world implementation. By incorporating conditions such as varying traffic densities, pedestrian demand, emergency vehicle presence, lane closures, road construction, and accidents, the study promotes a more proactive and data-driven approach to urban traffic management.

The findings of this study are expected to benefit the following:

- **Local Government Units (LGUs) and Traffic Management Authorities.** The system may serve as a simulation-based decision-support tool for evaluating traffic conditions, road constraints, and adaptive traffic signal strategies prior to actual implementation. This can assist traffic management offices in anticipating congestion buildup and assessing possible traffic responses under disruptive road conditions.
- **Traffic Planners and Urban Developers.** The simulation platform provides insights into traffic flow behavior, congestion patterns, and signal performance, supporting more informed planning and development of road networks, intersections, and traffic management strategies.
- **Traffic Enforcers.** Through simulated traffic scenarios and congestion analysis, traffic enforcers may gain a better understanding of traffic bottlenecks and road behavior, which can support more effective on-ground traffic management and coordination.
- **Drivers and Road Users.** The implementation of adaptive traffic signal optimization may help reduce unnecessary waiting times and improve overall traffic flow efficiency, contributing to a more convenient travel experience.

- **Pedestrians.** The system incorporates pedestrian behavior within the simulation environment, promoting safer and more balanced traffic signal management that considers pedestrian crossing activity alongside vehicle movement.
- **Researchers and Future Developers.** This study may serve as a reference and foundation for future research involving reinforcement learning, intelligent transportation systems, agent-based simulation, and AI-driven traffic management technologies.

Scope and Limitations

This study focuses on the development of SMARTFLOW, an AI-driven agent-based traffic simulation system designed to model and analyze traffic behavior within a selected intersection and its adjacent road segments in Tagum City. The system operates through a localized simulation environment developed using the Simulation of Urban Mobility (SUMO), where vehicles and pedestrians are represented as autonomous agents capable of dynamic interaction within the road network. Reinforcement Learning (RL) is implemented as one of the traffic signal control mechanisms, allowing the system to optimize signal timing decisions based on simulation-generated traffic conditions.

The simulation environment incorporates essential traffic components, including vehicle flow, pedestrian crossings, lane usage, traffic signal operations, and configurable traffic scenario parameters. These parameters include traffic density, pedestrian density, and emergency vehicle status. The system also includes configurable road constraint conditions, including constraint type, severity level, and affected lane. Road constraint scenarios may include lane closures, road construction, accidents, and temporary road blockages. These conditions are simulated to evaluate system adaptability and traffic signal performance under varying traffic and road disruption situations.

To improve realism, the study incorporates heterogeneous driver and pedestrian behaviors within the simulation environment. Vehicles and pedestrians are modeled with varying behavioral patterns, including aggressive driving tendencies, inconsistent lane discipline, delayed reaction behavior, and non-compliant pedestrian movement, to better approximate real-world urban traffic conditions. The study also utilizes field observations, traffic behavior analysis, and visual references from the selected area to approximate actual road geometry and traffic flow behavior within the simulation environment.

However, the system does not utilize live traffic feeds, CCTV processing, GPS tracking, or real-time sensor integration. The real-time metrics within the simulation environment generated by the system refer only to simulation-generated traffic data within the SUMO environment and do not represent live road conditions. The simulation environment serves only as a real-world-inspired approximation intended for traffic analysis and evaluation purposes.

The study is limited to a simulation-based implementation and does not involve deployment within actual traffic infrastructure. The system is not connected to physical traffic signals, Internet of Things (IoT) devices, or live traffic management systems. Additionally, the study is limited to a localized road network surrounding a selected intersection and does not represent a city-wide traffic model. While the system includes configurable traffic scenarios, selected road constraint conditions, and varying behavioral patterns, it does not fully account for all real-world variables such as extreme weather conditions, highly unpredictable human decisions, large-scale traffic network interactions, or complex multi-city traffic dynamics.

Despite these limitations, the study aims to demonstrate the potential of integrating agent-based simulation and Reinforcement Learning as a simulation-based decision-support approach for adaptive traffic signal

optimization and traffic analysis under varying traffic scenario and road constraint configurations.

Review of Related Literature and Studies

Related Literature

Traditional Traffic Management Systems

Traditional traffic management primarily relies on fixed-time control and manual enforcement to regulate urban mobility. Fixed-time signal systems function based on preset timing plans developed from historical traffic volume studies [11]. While these systems are cost-effective and easy to maintain, they are inherently unable to accommodate stochastic traffic fluctuations, often leading to excessive delays and queue spillback during peak periods and inefficient green time allocation during varying demand conditions [12]. In many developing urban centers, the limitations of these static systems are mitigated through manual traffic enforcement, where officers override signal operations based on visual observation [13]. However, this reactive approach is prone to human error and lacks the data-driven precision required to optimize complex road networks [14].

Intelligent Transportation Systems (ITS) in the Philippine Context

Intelligent Transportation Systems (ITS) integrate information technology and data-driven strategies to improve traffic efficiency and road safety [15]. In the Philippines, ITS development has focused on smart city frameworks that combine automated monitoring with reactive congestion management [16]. Despite the widespread deployment of traffic signal systems in urban areas, most implementations rely on fixed-time counters that change lights sequentially regardless of actual traffic conditions in respective lanes. These predetermined timing systems fail to adapt to real-time traffic density variations, causing unnecessary delays and congestion even when some lanes have minimal or no traffic [17]. This highlights the critical need for adaptive traffic management

systems that can continuously monitor lane-specific conditions and provide centralized coordination to optimize traffic flow across interconnected intersections.

Integration of Artificial Intelligence in Traffic Management

The integration of Artificial Intelligence (AI) marks a shift from reactive to proactive traffic management. AI applications utilize machine learning and predictive analysis to transform chaotic vehicular patterns into organized flows [18]. By processing real-time traffic data from surveillance cameras and neighboring intersection information through distributed Multi-Agent Reinforcement Learning algorithms, AI-driven traffic control systems can optimize multiple objectives including reducing average vehicle queue lengths and minimizing waiting times at interconnected intersections [19]. This technological evolution provides the necessary foundation for more specialized applications, such as computer vision and reinforcement learning, to handle the dynamic variables of urban road networks.

Reinforcement Learning in Traffic Signal Control

Reinforcement Learning (RL) has emerged as a superior alternative to traditional control by allowing agents to learn optimal signal policies through continuous interaction with a simulated environment [20]. Unlike rule-based systems, RL controllers utilize reward systems to dynamically adjust signal phases based on current demand [21]. Global studies indicate that Deep Reinforcement Learning (DRL) architectures, when integrated with simulation platforms, can significantly reduced delays [22]. Countries successfully implementing RL in traffic simulations demonstrate that these systems are particularly effective at managing unpredictable variables, such as emergency vehicle presence and pedestrian surges [23].

Agent-Based Traffic Simulation and SUMO

Agent-based modeling (ABM) provides the microscopic detail necessary for realistic traffic simulation, treating every vehicle and pedestrian as an

autonomous entity with unique behaviors [24]. The Simulation of Urban Mobility (SUMO) is widely recognized in the literature as the industry-standard tool for testing RL-based controllers, owing to its high-fidelity modeling and the TraCI API, which enables real-time signal manipulation [25]. These simulations enable researchers to model "heterogeneous" traffic—including public utility vehicles (PUVs) and aggressive driving behaviors—which is essential for accurately representing the Philippine traffic landscape [26].

Related Studies and Systems

Dual-Targeting Deep Reinforcement Learning Traffic Signal Control System

A study by Kodama et al. [27] proposed a multi-agent traffic light control system based on a dual-targeting algorithm, which emphasizes the reinforcement of traffic optimization experience. The system utilizes a deep reinforcement learning (DRL) agent integrated with the Simulation of Urban Mobility (SUMO) to reduce waiting times at complex intersections.

The framework consists of a neural network handled by a Traffic Signal Control System (TSCS). As illustrated in the conceptual diagram in Fig. 1, the input state includes vehicle position (P_t), lane density (D_t^l), change in density (ΔD_t^l), and velocity (V_t^l), which are processed through multiple hidden layers (256 and 128 ReLU units) to output Q-values for action selection. The system operates through a closed-loop interaction where the SUMO environment provides traffic states, and the RL agent issues signal commands (actions) while receiving rewards based on the minimization of cumulative waiting time.

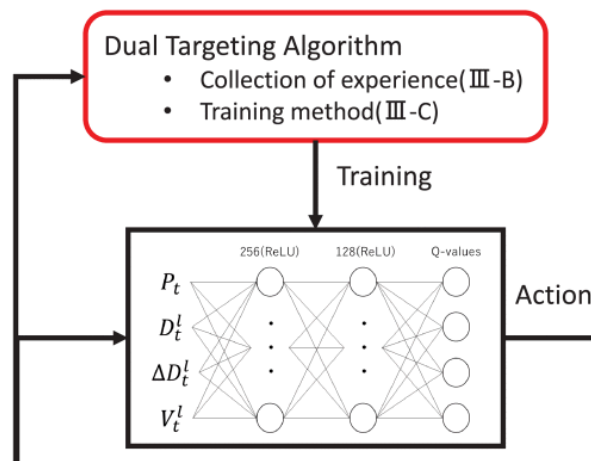
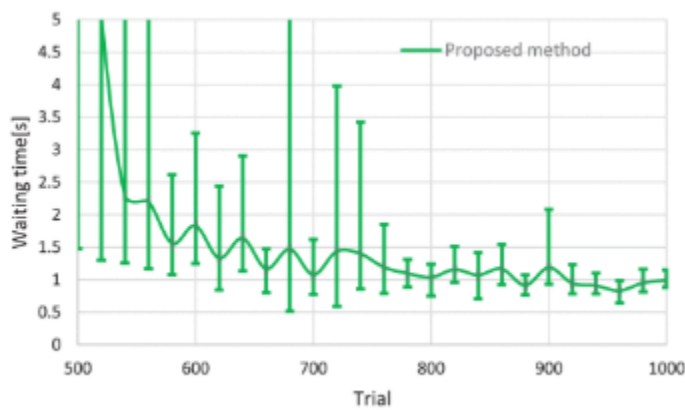


Fig. 1. Conceptual diagram of the proposed traffic light control system.

The effectiveness of the dual-targeting algorithm was validated through extensive simulation trials. The performance output, shown in Fig. 2, demonstrates the learning curves across 1,000 trials. While standard DQN and Multistep DQN architectures showed higher variance, the proposed "Dual-Targeting" method achieved a significantly lower unbiased sample variance of 0.02 at the 1,000th trial. This visual evidence confirms that the adaptive learning agent successfully stabilizes and minimizes vehicle waiting time as it gains more experience within the simulation.

Fig. 2. Sin



This syste

ing

L and the SUMO environment for adaptive signal optimization. A technical benchmark for state-space definitions (position and density) and reward functions based on "accumulated waiting time" [30]. However, SMARTFLOW expands on this foundation by introducing localized road constraint scenarios, such as lane closures and accidents and modeling heterogeneous pedestrian-vehicle

interactions to serve as a proactive decision-support tool for authorities in Tagum City.

TrafficSim: Learning to Simulate Realistic Multi-Agent Behaviors

A research study by Suo et al. [28] introduced TrafficSim, a multi-agent behavior model designed to generate realistic and socially consistent traffic simulations. Unlike traditional environments that rely on rigid, heuristic-based traffic rules, this system leverages real-world data to learn directly from human demonstrations, allowing it to capture naturalistic driving behaviors such as yielding, merging, and complex interactions.

The TrafficSim architecture utilizes an implicit latent variable model to jointly generate socially consistent plans for all actors in a scene. As shown in the architectural framework in Fig. 3, the system processes a bird's-eye view (BEV) representation of the map and actor states through a CNN-based encoder. It then employs a graph neural network (GNN) to model interactions between agents, followed by an MLP decoder that predicts long-term trajectories



Fig. 3. System architecture of TrafficSim illustrating the CNN encoder, interaction graph, and trajectory decoder.

The effectiveness of the TrafficSim system is demonstrated through its ability to generate socially consistent and diverse multi-agent trajectories. As illustrated in Fig. 4, the system generates multiple plausible "futures" (shown in red and blue) for each vehicle agent based on a single observation of the environment. When these predictions are compared to the actual ground truth (shown in green), the results confirm that the system accurately predicts

naturalistic driving behaviors, such as lane maintenance and collision avoidance, which are essential for high-fidelity urban simulations.

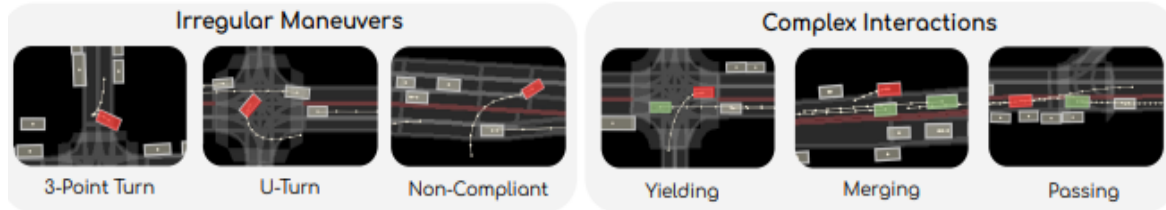


Fig. 4. Visual output of TrafficSim showing diverse multi-agent trajectory predictions and ground truth comparison.

This study is highly relevant to SMARTFLOW as it provides the theoretical and technical justification for modeling vehicles and pedestrians as autonomous agents with complex, non-linear behaviors. While TrafficSim focuses primarily on learning trajectories from real-world datasets to improve self-driving evaluation, SMARTFLOW adopts selected agent-based simulation principles to a localized Tagum City intersection to optimize traffic signal timing via Reinforcement Learning. By incorporating Suo et al.'s findings on "socially consistent" agent behavior, SMARTFLOW ensures that its simulation environment remains a high-fidelity "localized simulation environment" of the actual study site.

IoT-Based Adaptive Traffic Signal Controller for Congestion Reduction

A study by AlMulla et al. [29] developed an adaptable traffic light control system designed to improve flow at intersections by adjusting signal timing in real time. The research focuses on moving away from rigid, pre-timed controllers toward a system that perceives actual road demand through integrated sensors.

The system's architecture relies on an Internet of Things (IoT) framework that utilizes ultrasonic sensors positioned at specific distances along the road to detect vehicle presence and measure queue length. As shown in the architectural framework in Fig. 5, data from these sensors is processed by a

central controller that applies a smart algorithm to modify the default 60-second green time. The controller dynamically extends or reduces the phase duration based on the detected congestion level, ensuring that the intersection remains responsive to fluctuating traffic volumes.

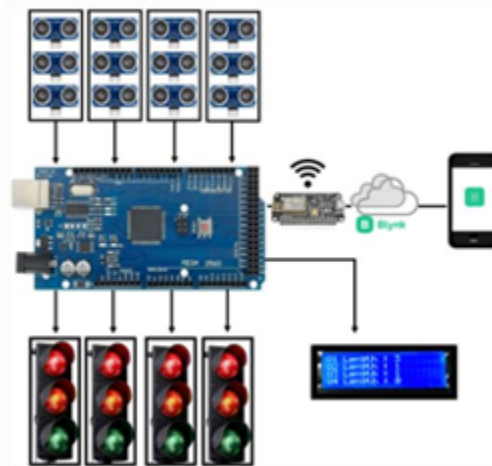


Fig. 5. System architecture of the IoT-based adaptive traffic signal controller.

The system's performance was validated through a working prototype and an accompanying mobile application. As illustrated in Fig. 6, the system provides a real-time visualization of congestion levels (e.g., Low, Medium, High) through a smartphone interface. The results of the study confirmed that the adaptive approach not only improved overall traffic flow but also significantly reduced the average wait time for drivers at the intersection.

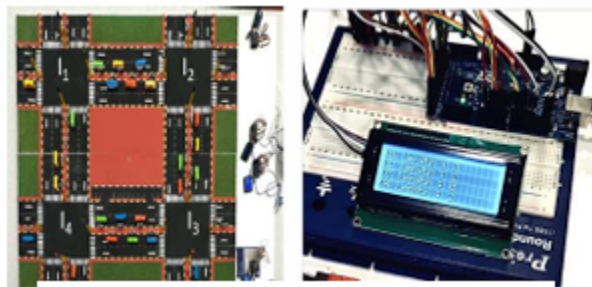


Fig. 6. Sample output showing the real-time congestion monitoring application.

This study is highly relevant to SMARTFLOW as it demonstrates the practical application of queue-length detection for signal optimization. Like AIMulla et al., SMARTFLOW seeks to replace inefficient fixed-time systems with adaptive logic. However, while this existing system focuses on a physical IoT sensor deployment, SMARTFLOW utilizes Simulation of Urban Mobility (SUMO) to model these same adaptive principles under localized road constraints—such as accidents or roadwork—allowing for a more diverse range of "what-if" testing without the high cost of physical hardware installation.

Synthesis of Related Studies and Systems

The reviewed studies demonstrate that Artificial Intelligence, Reinforcement Learning, and agent-based simulation have become effective approaches for adaptive traffic management and congestion reduction. Existing systems successfully implement dynamic traffic signal optimization, multi-agent interaction modeling, and intelligent traffic analysis using simulation environments such as SUMO and machine learning frameworks.

However, most existing studies primarily focus on generalized urban simulations, real-time sensor integration, or trajectory prediction models, with limited emphasis on localized traffic environments and road constraint analysis within the Philippine context. Additionally, many systems either rely heavily on IoT infrastructure or focus solely on vehicle optimization without incorporating heterogeneous pedestrian behavior and localized disruptive traffic scenarios such as lane closures, flooding, and road construction.

To address these gaps, SMARTFLOW proposes a localized AI-driven agent-based traffic simulation system that combines Reinforcement Learning–based traffic signal optimization with heterogeneous vehicle-pedestrian interaction and road constraint simulation. Unlike existing systems, the study

focuses on a selected intersection its adjacent road segments in Tagum City and aims to provide a simulation-based decision-support platform capable of evaluating adaptive traffic responses under varying traffic and road disruption conditions.

Table 1.Comparative Analysis of Related Traffic Management Systems and SMARTFLOW

Features	Dual-Targeting DRL [27]	TrafficSim [28]	IoT-Based Adaptive Traffic Signal Controller [29]	SMARTFLOW
Main Technology	Deep Reinforcement Learning	Multi-Agent Trajectory Learning	IoT + Adaptive Signal Control	RL + Agent-Base Simulation
Simulation Environment	SUMO	Learned Traffic Environment	Prototype-Based Traffic System	SUMO
Adaptive Signal Control	✓		✓	✓
Autonomous Agents	Partial	✓		✓
Pedestrian Modeling		Partial		✓

Road Constraints Analysis				✓
Emergency Vehicle Handling			Partial	✓
Real-Time Infrastructure Dependency			✓	
Localized Traffic Simulation				✓

Table 1 presents the comparative analysis of existing systems and the proposed SMARTFLOW Traffic System. The comparison shows that existing systems have contributed important features to traffic management, such as deep reinforcement learning for adaptive signal control, multi-agent behavior modeling, and IoT-based traffic monitoring. However, each existing system also has limitations when compared to the objectives of SMARTFLOW. The Dual-Targeting DRL system provides adaptive signal control using SUMO and reinforcement learning, but it does not fully address pedestrian modeling, emergency vehicle handling, road constraint analysis, or localized traffic simulation. TrafficSim focuses on autonomous multi-agent behavior and trajectory learning, but it does not directly support adaptive traffic signal control or road constraint analysis. Meanwhile, the IoT-Based Adaptive Traffic Signal Controller supports adaptive signal control and partial emergency vehicle handling, but it depends on physical infrastructure and does not provide a simulation-based environment for testing different traffic disruption scenarios.

In contrast, SMARTFLOW integrates reinforcement learning, agent-based simulation, pedestrian modeling, emergency vehicle handling, and localized road constraint analysis within a SUMO-based simulation environment. The table highlights that SMARTFLOW addresses the gaps found in the existing systems by providing a simulation-based decision-support platform that can evaluate traffic scenarios such as lane closures, road construction, accidents, pedestrian congestion, and emergency vehicle conditions without relying on real-time physical infrastructure. This makes SMARTFLOW more suitable for localized traffic analysis in the selected intersection in Tagum City, as it combines adaptive traffic signal optimization with realistic vehicle-pedestrian interaction and road disruption testing. Therefore, the comparison supports the need for the proposed system and justifies its development as a more comprehensive AI-driven traffic simulation and optimization tool.

Definition of Terms

Agent-Based Simulation. A simulation modeling approach in which individual entities, such as vehicles and pedestrians, are represented as autonomous agents capable of interacting within a virtual environment.

Artificial Intelligence (AI). The capability of computer systems to perform tasks that typically require human intelligence, such as learning, decision-making, and adaptive problem-solving.

Autonomous Agent. An independent entity capable of perceiving its environment and making decisions or performing actions without direct human intervention.

Decision-Support System. A computer-based system designed to assist users in analyzing scenarios and making informed decisions through data analysis and simulation.

Emergency Vehicle Priority. A traffic management mechanism that adjusts signal timing to provide faster passage for emergency vehicles during traffic operations.

Heterogeneous Agent Behavior. The variation of behavioral patterns among simulated agents includes differences in driving style, pedestrian movement, and rule compliance.

Intersection Traffic Flow. The movement and interaction of vehicles and pedestrians within a signalized road intersection.

Queue Length. The number of vehicles waiting at a traffic signal or congested roadway segment.

Reinforcement Learning (RL). A machine learning technique in which an agent learns optimal actions by interacting with an environment and receiving rewards or penalties based on its decisions.

Road Constraint. A condition or obstruction that affects normal traffic flow, such as lane closures, road construction, accidents, flooding, or temporary blockages.

Simulation of Urban Mobility (SUMO). An open-source microscopic traffic simulation software used for modeling and analyzing vehicle and pedestrian movement within road networks.

Traffic Signal Control. The management and adjustment of traffic light operations to regulate vehicle and pedestrian movement at intersections.

Traffic Simulation. The use of computer-based models to replicate, analyze, and evaluate traffic behavior within a virtual environment.

Adaptive Traffic System. A traffic management system that dynamically adjusts traffic signal timing based on changing traffic conditions and traffic demand.

Pedestrian Demand. The volume or frequency of pedestrians requiring access to cross a specific roadway or intersection.

Traffic Density. The concentration of vehicles within a specific roadway segment or traffic area at a given time.

CHAPTER II

METHODOLOGY

The study adopts a Hybrid Development Methodology that combines the structured approach of the Waterfall model and the flexibility of Agile practices to effectively support the development of the SMARTFLOW system, as illustrated in Figure 7. The Waterfall approach is applied during the initial stages of the study, particularly in planning, requirements analysis, and system design. This ensures that the system objectives, simulation requirements, traffic modeling components, and system structure are clearly defined before development begins. Meanwhile, Agile practices are utilized during the implementation, testing, and refinement phases, allowing iterative development, continuous simulation, repeated testing, debugging, and incremental improvement of system components such as the traffic simulation environment, reinforcement learning-based traffic signal controller, and dashboard interface. This integration provides a balance between structure and adaptability, making it suitable for an AI-driven simulation system that involves both defined requirements and continuous experimentation and optimization.

The study begins with the Planning Phase, where the proponents define the project scope, objectives, timeline, and required resources based on the approved capstone proposal. During this phase, the proponents also identify the selected intersection and traffic scenarios to be used in the simulation environment and establish the overall direction and feasibility of the system.

In the Analysis Phase, the proponents gather and finalize the system requirements by analyzing traffic conditions, traffic flow behavior, pedestrian movement, and the requirements necessary for the development of the

simulation and reinforcement learning components. Functional and non-functional requirements are also identified during this phase.

In the Design Phase, the proponents translate the identified requirements into a structured system architecture, including the design of the simulation workflow, reinforcement learning framework, dashboard interface, and overall system interaction flow. Diagrams and system models are also prepared to visualize how the system will operate.

During the Implementation Phase, the proponents will develop the SMARTFLOW system using an iterative approach. The simulation environment, reinforcement learning integration, and dashboard interface will be built, tested, and improved through repeated development cycles. The proponents will configure the SUMO simulation environment, model vehicle and pedestrian agents, integrate Python and TraCI communication, implement the reinforcement learning model, and develop the local dashboard and SQLite database. Since the system has not yet been launched, this phase will focus on preparing the core system components for testing and evaluation. Each iteration will allow the proponents to identify issues, apply improvements, and refine the system before proceeding to the final testing and evaluation phase.

In the Testing Phase, the proponents conduct continuous simulation testing to evaluate system behavior under different traffic scenarios and road constraint conditions. This includes testing the simulation environment, traffic signal responses, reinforcement learning performance, and dashboard outputs.

Finally, in the Refinement Phase, the proponents conduct system refinement activities by identifying detected issues and applying necessary debugging, adjustments, and optimizations to improve system performance, reliability, and usability. This phase includes refining simulation parameters, improving reinforcement learning performance, correcting identified system issues, and enhancing the overall efficiency of the SMARTFLOW system. Although limited within the duration of the study, this phase ensures that the

system remains stable, functional, and suitable for future enhancements and extended simulation scenarios.

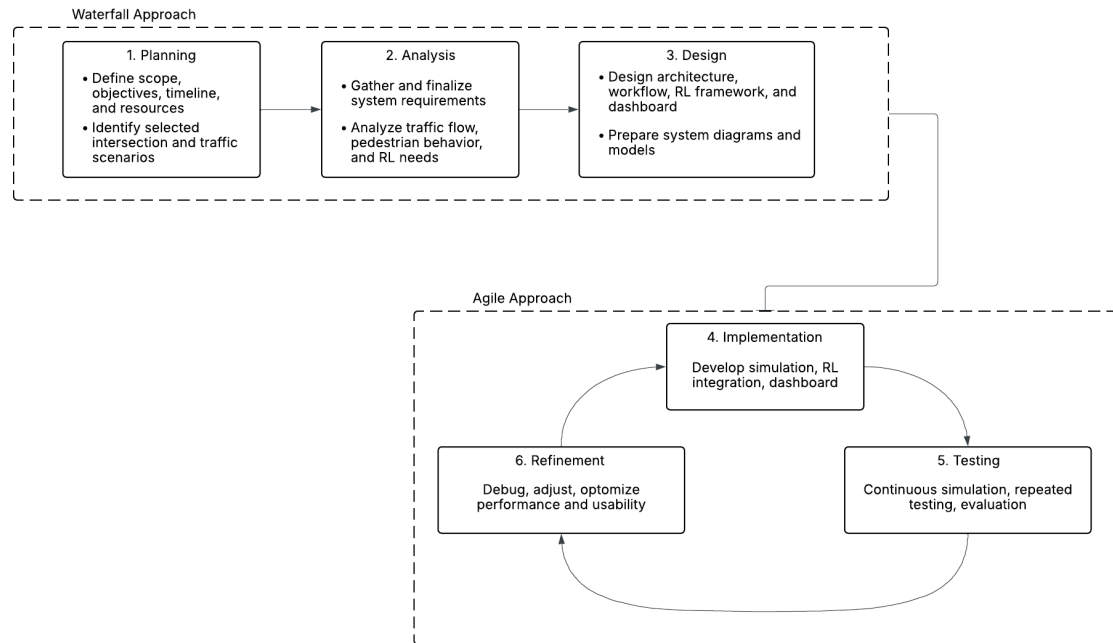


Figure 7. Hybrid Development Methodology

SYSTEM PLANNING

Project Team Organization

The project team, as shown in Figure 8, is organized to ensure effective coordination, clear communication, and proper distribution of responsibilities throughout the development of the SMARTFLOW system. The Adviser provides academic and technical guidance to ensure that the study meets the required research standards. The Project Manager and Documentation lead oversees project planning, task coordination, progress monitoring, timeline management, and the preparation of project documentation. Under this role are the System Developer and the System Analyst. The System Developer is responsible for

coding, system implementation, simulation development, and technical integration. Meanwhile, the System Analyst is responsible for analyzing system requirements, identifying system processes, and supporting the design and evaluation of the SMARTFLOW system. This structure allows the team to maintain balanced coordination between academic guidance, project management, documentation, technical development, and system analysis.

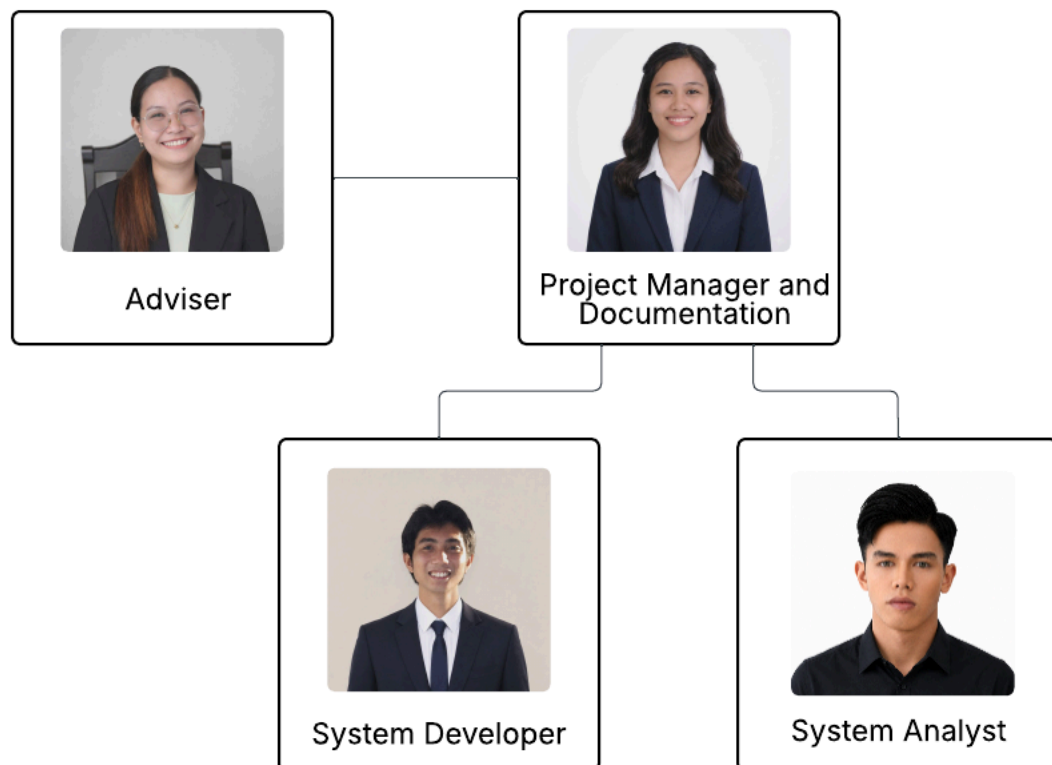


Figure 8. Project Team Organization

Work Breakdown Structure

The Work Breakdown Structure shown in Figure 9 presents the phase-based breakdown of the SMARTFLOW Traffic System development process. It follows the Hybrid Development Methodology adopted in the study, where the early phases follow a structured Waterfall approach and the later

phases follow an Agile approach through continuous testing, refinement, and improvement. The WBS organizes the project into six major phases: Planning, Analysis, Design, Implementation, Testing and Evaluation, and Refinement.

During the Planning Phase, the proponents define the project scope and objectives, identify the project timeline and required resources, select the target intersection in Tagum City, and identify the traffic scenarios and road constraint conditions to be included in the simulation. This phase establishes the foundation of the project by clarifying what the system aims to develop and what traffic situations it needs to represent.

The Analysis Phase focuses on understanding the traffic environment and the system requirements. This includes analyzing traffic flow behavior, pedestrian movement, and the requirements needed for the simulation and reinforcement learning components. In this phase, the proponents also identify the functional and non-functional requirements of the system and define the reinforcement learning requirements needed for adaptive traffic signal control.

In the Design Phase, the proponents prepare the structure and models of the SMARTFLOW system. This includes designing the system architecture, simulation workflow, reinforcement learning framework, dashboard interface, and system diagrams. These design outputs serve as the guide for how the system components will interact during development and simulation execution.

During the Implementation Phase, the proponents will develop the SMARTFLOW system using an iterative approach. The simulation environment, reinforcement learning integration, and dashboard interface will be built, tested, and improved through repeated development cycles. The proponents will configure the SUMO simulation environment, model vehicle and pedestrian agents, integrate Python and TraCI communication, implement the reinforcement learning model, and develop the local dashboard and SQLite database. Since the system has not yet been launched, this phase will focus on preparing the core system components for testing and evaluation. Each iteration will allow the

proponents to identify issues, apply improvements, and refine the system before proceeding to the final testing and evaluation phase.

The Testing and Evaluation Phase involves running traffic and road constraint scenarios to assess the behavior and performance of the system. This phase includes testing fixed-time and reinforcement learning–based signal control, collecting performance metrics, and comparing and analyzing the results. The evaluation focuses on determining how effectively the system supports traffic simulation and adaptive signal optimization.

Finally, the Refinement Phase focuses on improving the system based on the results of testing and evaluation. This includes debugging system issues, adjusting simulation parameters, improving reinforcement learning performance, and finalizing system outputs and reports. This phase ensures that the system is made more stable, usable, and suitable for presentation, evaluation, and future enhancement.

Overall, the Work Breakdown Structure provides a clear and organized guide for managing the development of the SMARTFLOW Traffic System. It ensures that each major task is properly grouped, sequenced, and aligned with the hybrid methodology used in the study.

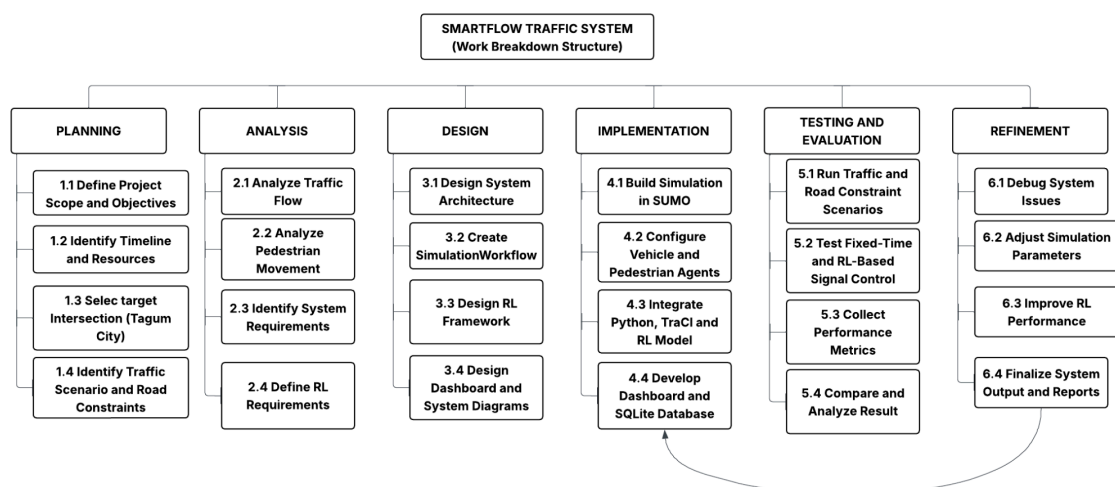


Figure 9. Work Breakdown Structure

Gantt Chart

The Gantt chart shown in Figure 10 presents the timeline and schedule of activities for the development of the SMARTFLOW Traffic System. It is aligned with the Hybrid Development Methodology and the Work Breakdown Structure adopted in the study. The chart begins with the structured Waterfall phases of Planning, Analysis, and Design, followed by the Agile-based iterative phases of Implementation, Testing and Evaluation, and Refinement.

The Planning Phase covers the definition of the project scope and objectives, identification of timeline and resources, selection of the target intersection, and identification of traffic scenario parameters and road constraint conditions. The Analysis Phase focuses on analyzing traffic flow, pedestrian movement, system requirements, and reinforcement learning requirements. The Design Phase includes the preparation of the system architecture, simulation workflow, reinforcement learning framework, dashboard interface, and system diagrams.

The Implementation, Testing and Evaluation, and Refinement phases are presented as overlapping and iterative activities. During implementation, the proponents build the SUMO simulation environment, model vehicle and pedestrian agents, integrate Python, TraCI, and the reinforcement learning model, and develop the dashboard and SQLite database. During testing and evaluation, the proponents run traffic and road constraint scenarios, test fixed-time and reinforcement learning–based signal control, collect performance metrics, and compare and analyze the results. During refinement, the proponents debug system issues, adjust simulation parameters, improve reinforcement learning performance, and finalize system outputs and reports.

The Gantt chart also shows that the iterative phases may repeat depending on testing results and system improvements. This reflects the Agile component of the methodology, where issues found during testing are addressed through refinement and may return to implementation for correction or enhancement. Overall, the Gantt chart supports the organized and timely completion of the SMARTFLOW Traffic System while remaining consistent with the six major phases of the Work Breakdown Structure.

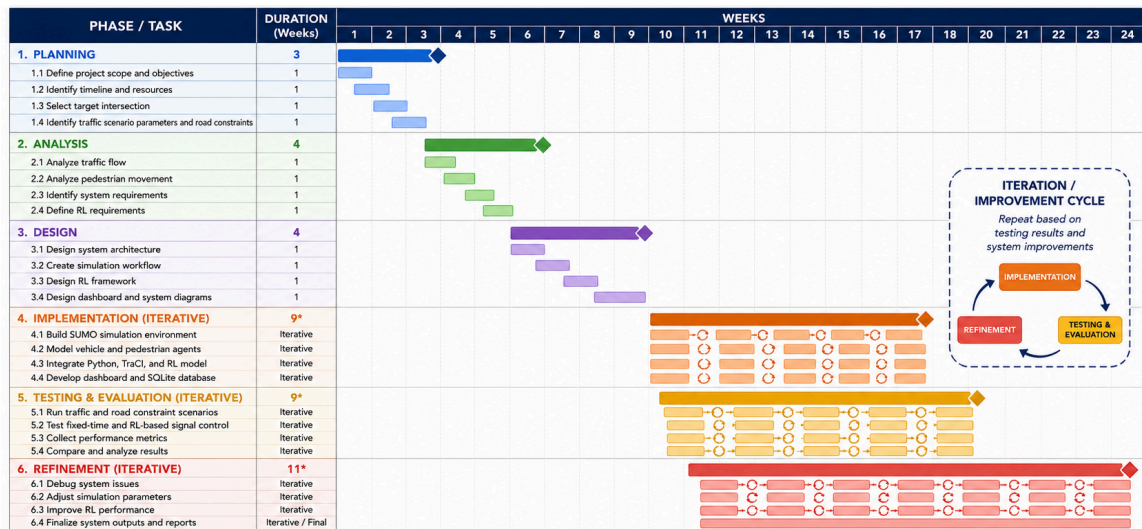


Figure 10. Gantt Chart

SYSTEM ANALYSIS

System Architecture

The system architecture of the SMARTFLOW Traffic System, as illustrated in Figure 11, presents a structured framework that demonstrates how the different components of the system interact to simulate, analyze, store, and present traffic data under varying traffic and road constraint scenarios using reinforcement learning. The system is composed of six major layers: the User/Client Layer, Simulation Layer, Control and Communication Layer, Intelligence Layer, Database Layer, and Outputs Layer.

The User Layer represents the authorized operator, such as a researcher or CTTMO personnel, who interacts with the system through the local dashboard. Through this layer, the operator can log in, configure traffic scenarios, set road constraints, select the control mode, run or stop the simulation, monitor real-time traffic metrics within the simulation, view performance reports, and log out of the system.

The Simulation Layer utilizes the Simulation of Urban Mobility (SUMO) to model the localized traffic environment. This layer includes road network modeling, vehicle simulation, pedestrian simulation, and road constraint configuration. It represents the selected intersection, lanes, links, traffic signals, vehicle movement, pedestrian crossings, and disruption scenarios such as lane closures, road construction, accidents, and emergency vehicle conditions.

The Control and Communication Layer serves as the communication bridge between the simulation environment and the intelligent decision-making component. This layer uses Python and TraCI to retrieve real-time traffic data from SUMO, control simulation execution, and send traffic signal control actions back to the SUMO environment.

The Intelligence Layer contains the Reinforcement Learning Engine implemented using PyTorch. This layer observes the current traffic state, selects signal actions, computes rewards, and learns an optimal policy for adaptive signal control. It supports traffic signal optimization by aiming to reduce congestion, minimize waiting time, improve traffic flow efficiency, and prioritize emergency and pedestrian movement when necessary.

The Database Layer uses SQLite as a lightweight local database for storing and managing system data. It stores user accounts, traffic scenarios, road constraints, signal control modes, simulation runs, traffic metrics, reinforcement learning outputs, and performance reports. This allows the system to organize

local data without requiring a centralized database server or cloud-based storage.

The Outputs Layer presents the results generated by the system. These outputs include real-time traffic metrics within the simulation, queue length analysis, waiting time analysis, traffic reports, reinforcement learning versus fixed-time signal control comparison, and dashboard visualizations. Through these outputs, the operator can evaluate the performance of the SMARTFLOW system under different traffic scenarios and road constraint conditions.

Overall, the system architecture demonstrates an integrated simulation-based intelligent traffic management environment where the operator configures scenarios, SUMO executes the traffic simulation, Python and TraCI manage communication, the reinforcement learning engine determines adaptive signal actions, SQLite stores the generated data, and the dashboard presents reports and visualizations for traffic analysis and decision support.

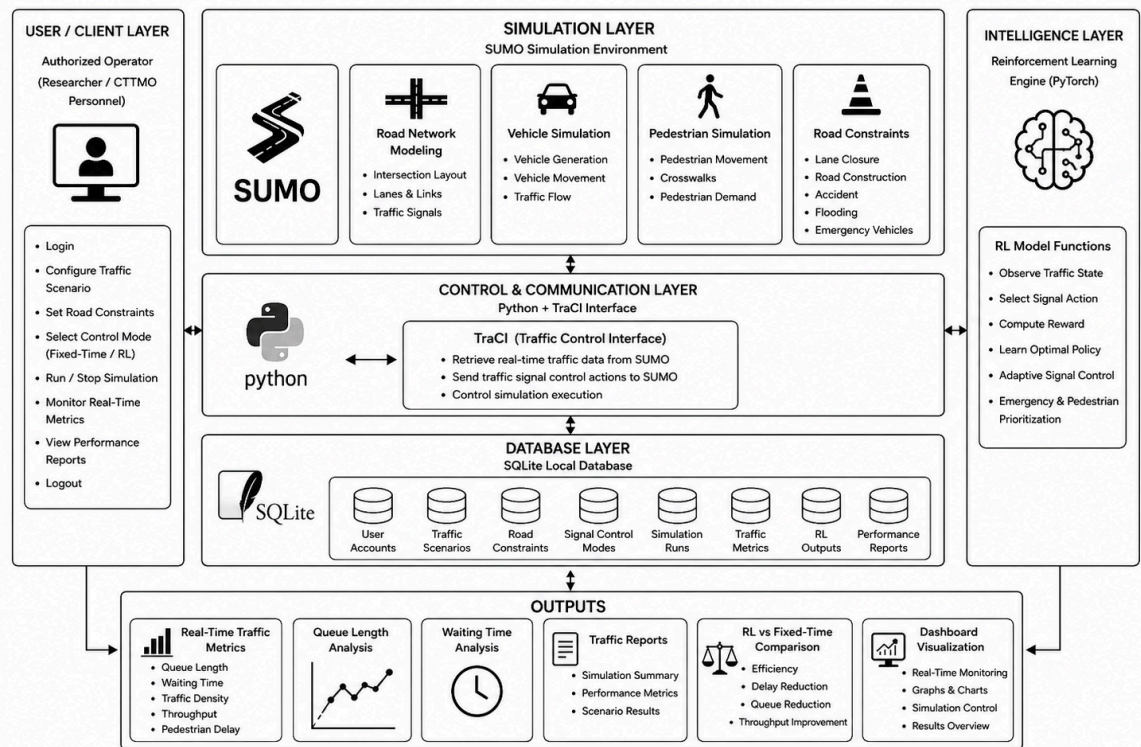


Figure 11. System Architecture Diagram

Conceptual Framework

Figure 12 shows the conceptual framework of the SMARTFLOW Traffic System. The framework presents how the system receives configuration inputs from the Authorized Operator, processes these inputs through a SUMO-based simulation environment and reinforcement learning–based traffic signal control, and produces simulation-generated monitoring results and performance reports through a local dashboard. Based on the requirements of researchers and traffic management personnel, the framework ensures that the system provides useful tools for configuring traffic scenarios, applying road constraint conditions, running simulations, monitoring traffic behavior, and comparing reinforcement learning–based signal control with fixed-time traffic signal control.

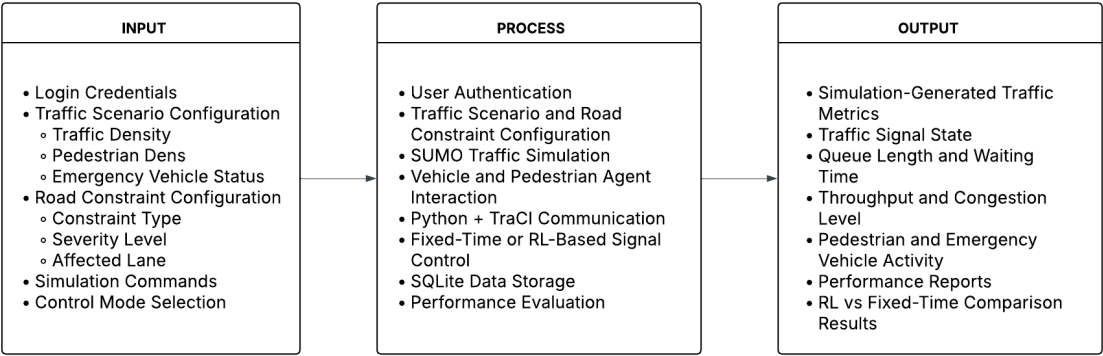


Figure 12. Conceptual Framework

Input

The input of the system includes the data and configuration settings provided by the Authorized Operator. These inputs include login credentials, traffic scenario configuration, road constraint configuration, simulation commands, and control mode selection. The traffic scenario configuration includes traffic density, pedestrian density, and emergency vehicle status. The

road constraint configuration includes the constraint type, severity level, and affected lane. The system also receives simulation commands such as initialize, start, pause, stop, reset, and reconfigure simulation. In addition, the operator selects the control mode, either fixed-time traffic signal control or reinforcement learning–based traffic signal control. These inputs are stored and managed using the local SQLite database before being used during simulation execution.

Process

The process begins when the Authorized Operator logs in to the system through the local dashboard. After authentication, the operator configures the traffic scenario and road constraint settings. The system then initializes the SUMO simulation environment based on the configured traffic density, pedestrian density, emergency vehicle status, constraint type, severity level, and affected lane. Vehicles and pedestrians are simulated as autonomous agents that interact within the selected intersection and adjacent road segments.

During simulation runtime, Python and TraCI serve as the communication layer between SUMO and the signal control component. The system collects simulation-generated traffic data such as queue length, waiting time, vehicle count, pedestrian count, traffic density, and signal state. If fixed-time control is selected, the system applies predefined signal timings for baseline comparison. If reinforcement learning–based control is selected, the RL controller analyzes the current simulation-generated traffic state, selects adaptive signal actions, computes reward values, and updates signal decisions to improve traffic flow. The generated traffic metrics, signal decisions, RL outputs, and simulation records are stored in the SQLite database for monitoring, evaluation, and reporting.

Output

The output of the system is a local dashboard that presents simulation-generated traffic monitoring results and performance reports. The

dashboard displays simulation status, traffic metrics, queue length, waiting time, throughput, congestion level, pedestrian activity, emergency vehicle activity, road constraint status, and traffic signal state. The system also generates performance reports and comparison results between reinforcement learning–based signal control and fixed-time traffic signal control. These outputs help researchers and traffic management personnel evaluate traffic behavior, assess the impact of road constraints, and support simulation-based decision-making for adaptive traffic signal optimization.

Functional and Non-functional Requirements

The functional and non-functional requirements define the expected capabilities and quality attributes of the proposed SMARTFLOW Traffic System. The functional requirements describe the core features of the system, including traffic simulation, road constraint scenario management, reinforcement learning–based traffic signal optimization, real-time monitoring within the simulation, and performance evaluation. Meanwhile, the non-functional requirements describe the operational quality of the system based on characteristics such as performance efficiency, reliability, usability, maintainability, and adaptability aligned with ISO/IEC 25010 software quality standards. These requirements ensure that the system operates efficiently, produces accurate simulation results, and supports effective traffic analysis and decision-making.

Functional Requirements

The system shall provide the following functionalities:

1. The system shall allow the user to configure traffic scenario parameters, including traffic density, pedestrian density, and emergency vehicle status.
2. The system shall allow the user to configure road constraint conditions, including constraint type, severity level, and affected lane. Constraint types may include lane closures, road construction, accidents, and

temporary road blockages.

3. The system shall initialize and execute a traffic simulation environment using the Simulation of Urban Mobility (SUMO).
4. The system shall simulate the movement and interaction of vehicles and pedestrians within the selected road network and configured traffic conditions.
5. The system shall collect simulation-generated traffic data, such as queue length, waiting time, vehicle count, pedestrian count, and traffic density, from the simulation environment during runtime.
6. The system shall process simulation data through a Python-based control layer integrated with SUMO using the Traffic Control Interface (TraCI).
7. The system shall implement a reinforcement learning–based traffic signal controller using PyTorch for adaptive signal optimization.
8. The system shall determine and apply traffic signal phases and timing adjustments based on the current simulation-generated traffic state and road conditions.
9. The system shall continuously update simulated traffic conditions and repeat the reinforcement learning decision-making process throughout the simulation runtime.
10. The system shall display simulation-generated traffic metrics, simulation status, signal states, and scenario information through a dashboard interface.
11. The system shall allow the user to start, pause, stop, reset, and reconfigure simulation scenarios through the dashboard interface.
12. The system shall store user credentials, simulation configurations, road constraint settings, traffic metrics, reinforcement learning outputs, and performance reports using SQLite as a lightweight local database, with optional export to CSV or JSON files.
13. The system shall generate simulation results and performance reports that can be used to evaluate and compare reinforcement learning–based traffic

signal optimization with fixed-time traffic control methods.

Non-functional Requirements

The following are the non-functional requirements of the system:

1. The system shall process simulation data and update traffic signal decisions in real time with minimal delay to ensure smooth and continuous simulation execution.
2. The system shall maintain stable simulation execution and consistent reinforcement learning performance with minimal system interruptions or processing errors.
3. The system shall provide a simple, organized, and user-friendly dashboard that enables users to easily configure scenarios, control simulations, and interpret traffic metrics.
4. The system shall ensure secure handling and storage of locally generated simulation data and configuration files within the local operating environment.
5. The system shall operate on standard desktop environments capable of supporting SUMO, Python, PyTorch, and related simulation tools.
6. The system shall follow a modular design structure to support easy updates, debugging, and enhancement of the simulation environment, reinforcement learning model, and dashboard components.
7. The system shall support future expansion to additional traffic intersections, road scenarios, traffic parameters, and simulation conditions without requiring major system redesign.
8. The system shall remain accessible locally without requiring continuous internet connectivity during simulation execution.

Use Case Diagram

The use case diagram, as illustrated in Figure 13, presents the interaction between the primary actor and the SMARTFLOW Traffic System. The system is designed as a localized simulation-based traffic analysis and decision-support platform accessible to authorized operators such as researchers and CTTMO personnel for traffic scenario configuration, road constraint testing, and reinforcement learning–based traffic signal optimization.

The primary actor identified in the system is the Authorized Operator, who is responsible for configuring, controlling, monitoring, and evaluating the simulation environment through a local dashboard interface. The dashboard serves as the main access point for managing simulation activities and analyzing traffic behavior under different traffic scenario and road constraint configurations.

The primary use cases of the system include user authentication, traffic scenario configuration, road constraint configuration, simulation management, traffic monitoring within the simulation, and system performance evaluation. Through the authentication feature, authorized personnel can securely access the system by entering locally managed login credentials, which are then validated before the user is allowed to operate the simulation platform.

The traffic scenario configuration functionality allows the Authorized Operator to define traffic conditions by setting parameters such as traffic density, pedestrian density, and emergency vehicle status. In addition, the road constraint configuration feature enables the operator to define road disruption conditions by specifying the constraint type, severity level, and affected lane. These configured conditions influence vehicle and pedestrian behavior and modify the operational state of the simulated road network during simulation execution.

Once the simulation is initiated, the system executes the traffic environment within the SUMO simulation platform and processes simulation-generated traffic conditions based on the selected control mode. The operator may select either fixed-time control or reinforcement learning–based

control. When reinforcement learning–based control is selected, the system applies adaptive signal decisions dynamically to optimize traffic flow efficiency and reduce congestion. When fixed-time control is selected, the system applies predefined signal timing for comparison and baseline evaluation.

Throughout the simulation process, the Authorized Operator can monitor traffic conditions and evaluate system performance through the dashboard interface. The system displays simulation outputs such as queue lengths, average delay, traffic throughput, traffic signal states, congestion levels, pedestrian activity, and simulation-generated performance metrics. The operator may also initialize, start, pause, stop, reset, and reconfigure the simulation for further testing and analysis.

Overall, the use case diagram demonstrates how authorized users interact with the SMARTFLOW Traffic System to configure traffic scenarios, manage road constraints, control simulation execution, monitor traffic behavior, and evaluate the effectiveness of reinforcement learning–based traffic signal optimization compared with fixed-time traffic signal control within a simulated traffic environment.

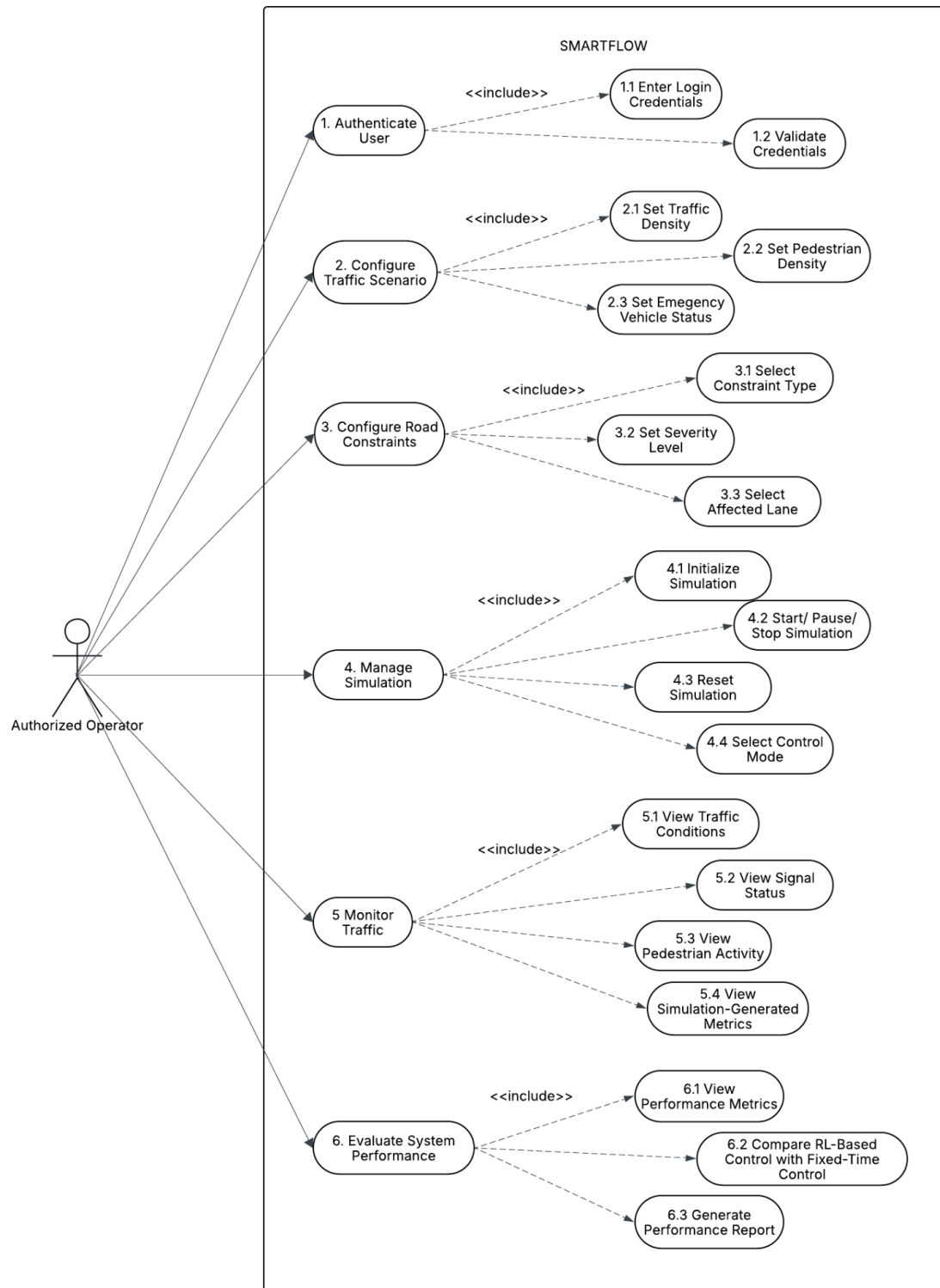


Figure 13. Use Case Diagram

Context Flow Diagram

The Context Flow Diagram shown in Figure 14 presents a high-level view of the SMARTFLOW Traffic System, illustrating how the system interacts with the external entity and how data flows across the simulation-based platform. It focuses on the logical flow of system inputs and outputs without detailing the internal processes of the system.

In the diagram, the main external entity is the Authorized Operator, which may include researchers, traffic management personnel, or CTTMO staff. The Authorized Operator provides the main inputs to the system, such as login credentials, traffic scenario configuration, road constraint configuration, simulation commands, and fixed-time or reinforcement learning–based control mode selection. The traffic scenario configuration includes parameters such as traffic density, pedestrian density, and emergency vehicle status, while the road constraint configuration includes the constraint type, severity level, and affected lane. These inputs allow the system to initialize and execute the simulation environment based on the configured traffic and road conditions.

The SMARTFLOW Traffic System processes the provided inputs by executing the traffic simulation, generating vehicle and pedestrian movements, applying road constraint conditions, and controlling traffic signals based on the selected control mode. When reinforcement learning mode is selected, the system uses the trained RL controller to determine adaptive signal actions. When fixed-time mode is selected, the system applies predefined traffic signal timing for baseline comparison.

The system produces outputs for the Authorized Operator through the dashboard interface. These outputs include real-time traffic metrics and status within the simulation, traffic signal states, queue lengths, waiting times,

throughput, congestion levels, pedestrian activity, simulation logs, and performance reports. The generated results support the evaluation and comparison of reinforcement learning–based adaptive traffic signal control against fixed-time traffic signal control under different traffic scenario and road constraint configurations.

The diagram also shows key data flows such as user inputs, traffic scenario configurations, road constraint configurations, simulation commands, traffic performance metrics, and generated reports. These flows demonstrate how input data is processed by the SMARTFLOW system and transformed into meaningful outputs that support traffic analysis, decision-making, and simulation-based evaluation.

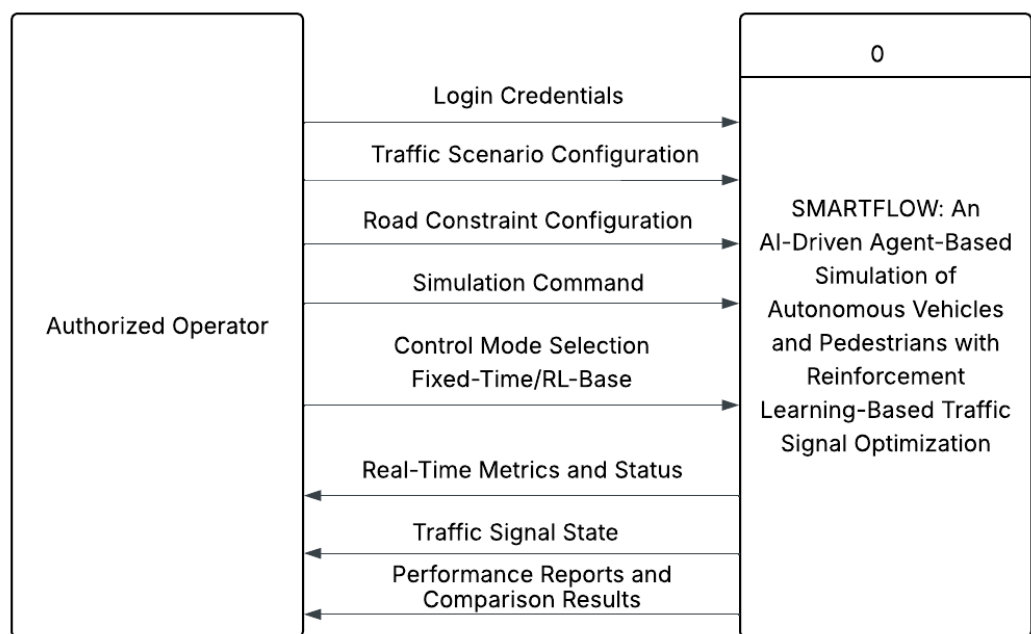


Figure 14. Context flow diagram of the system.

Data Flow Diagram

The Data Flow Diagram shown in Figure 15, the system is decomposed into five major processes: User Account Authentication, Traffic Scenario and Road Constraint Configuration, Simulation and Data Collection, Signal Control Processing, and Monitoring and Reporting. The Authorized Operator, which may include researchers or CTTMO personnel, serves as the primary external entity that interacts with the system. The operator provides login credentials, traffic scenario configuration, road constraint configuration, simulation commands, and control mode selection.

The User Account Authentication process validates the login credentials provided by the Authorized Operator. To perform this validation, the process sends a user account query to the SQLite database and retrieves user account records. After the credentials are checked, only authorized users are allowed to access and operate the SMARTFLOW simulation system.

After authentication, the Traffic Scenario and Road Constraint Configuration process manages the traffic scenario and road constraint inputs provided by the Authorized Operator. The traffic scenario configuration includes traffic density, pedestrian density, and emergency vehicle status, while the road constraint configuration includes the constraint type, severity level, and affected lane. These configured scenario and road constraint settings are stored in the SQLite database as part of the system configuration. The saved scenario configurations can then be retrieved and used by the system during simulation execution.

The Simulation and Data Collection process receives simulation commands from the Authorized Operator, such as starting, pausing, stopping, resetting, or running the simulation. This process retrieves the traffic scenario configuration, road constraint settings, and simulation parameters from the SQLite database. Based on these data, the system executes the SUMO-based traffic simulation and collects simulation-generated traffic information, such as vehicle count, pedestrian activity, queue length, waiting time, traffic density, and

traffic flow conditions. The collected simulation data and traffic metrics are then stored in the database for further processing and evaluation.

The Signal Control Processing process handles the control mode selected by the Authorized Operator. This process determines whether the system will use fixed-time signal control or reinforcement learning–based signal control. It retrieves traffic metrics and signal control parameters from the SQLite database, analyzes the current simulation-generated traffic conditions, and generates signal decisions. In reinforcement learning mode, the process also produces reward data used to evaluate and improve signal control decisions. These outputs are stored in the database and later used for monitoring, reporting, and performance comparison.

The Monitoring and Reporting process retrieves simulation results, reinforcement learning outputs, and historical reports from the SQLite database. This process presents the system outputs to the Authorized Operator through the dashboard interface. The main outputs include real-time metrics and status within the simulation, traffic signal state, and performance reports with comparison results. These outputs allow the operator to monitor the simulation, observe signal behavior, and evaluate the performance of reinforcement learning–based traffic signal control compared with fixed-time traffic signal control.

Overall, the Level 1 Data Flow Diagram demonstrates how SMARTFLOW manages user inputs, traffic scenario configuration, road constraint configuration, simulation execution, signal control processing, database storage, and reporting through a centralized SQLite database. By matching the number and meaning of the external data flows shown in the Context Flow Diagram, the DFD provides a clearer and more consistent representation of how data flows through the SMARTFLOW Traffic System.

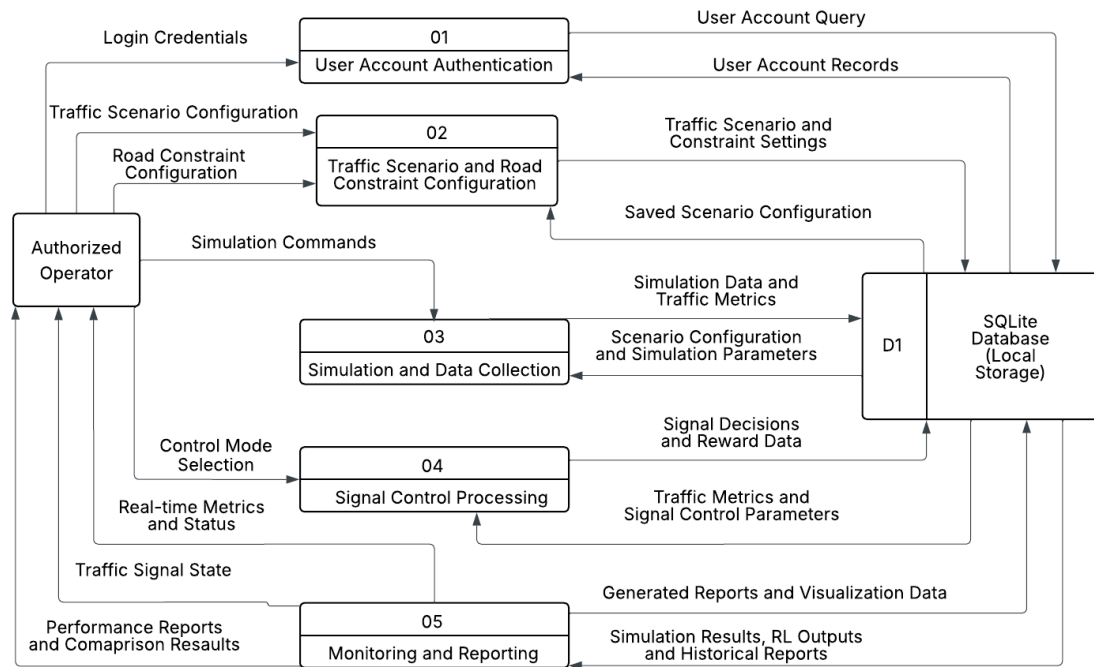


Figure 15. Data flow diagram of the system.

Traffic Scenario Configuration

Pedestrian Density Configuration

SYSTEM DESIGN

Entity Relationship Diagram

The Entity Relationship Diagram illustrated in Figure 16 presents the database structure of the SMARTFLOW Traffic System using SQLite as a lightweight local database. Since the system is designed as a localized simulation-based platform, SQLite is used to store and manage essential system data without requiring a centralized database server or cloud-based storage. The database supports the organization of user account credentials, traffic scenarios, road constraint configurations, signal control modes, simulation runs, traffic metrics, reinforcement learning outputs, and performance reports.

The User Accounts entity stores the login credentials used to access the system dashboard. Since the system is designed as a localized prototype, role-based access control is not included in the current scope, and the account record is mainly used for authentication. The Traffic Scenarios entity contains predefined traffic conditions such as normal traffic, heavy congestion, pedestrian-intensive conditions, and emergency traffic scenarios. The Road Constraints entity stores disruption settings such as lane closures, road construction, accidents, and pedestrian congestion, which may be applied during simulation execution.

The Signal Control Modes entity identifies the control strategy used in each simulation run, such as fixed-time control or reinforcement learning–based adaptive control. The Simulation Runs entity serves as the central table of the database because it records each executed simulation and links the selected user account, traffic scenario, road constraint, and signal control mode. This entity also stores important execution details such as the start time, end time, and simulation status.

The Traffic Metrics entity records the summarized performance data generated after each simulation run, including queue length, waiting time, throughput, pedestrian delay, traffic density, and signal efficiency. Since each simulation run produces one final set of summarized traffic metrics, the relationship between Simulation Runs and Traffic Metrics is one-to-one. The Performance Reports entity contains the summarized results and comparison output of each simulation run. Since each simulation run produces one final performance report, the relationship between Simulation Runs and Performance Reports is also one-to-one. Meanwhile, the RL Outputs entity stores reinforcement learning–related data such as traffic state information, selected signal actions, reward values, and timestamps. Since the reinforcement learning model may generate multiple actions and rewards during a single simulation run, one simulation run may have multiple RL output records.

The relationships between the entities are represented using Crow's Foot notation. One user account may be associated with multiple simulation runs, while each simulation run is linked to one user account. Similarly, one traffic scenario, one road constraint, and one signal control mode may be used in multiple simulation runs, but each simulation run uses only one selected record from each of these entities. A simulation run may generate multiple traffic metric records and reinforcement learning outputs, which may then be used to produce performance reports.

Overall, the ERD shows how SMARTFLOW organizes and connects the data needed for simulation execution, reinforcement learning processing, and performance evaluation. This database structure allows the system to store simulation data locally, maintain organized records, and support comparison between adaptive and fixed-time traffic signal control strategies.

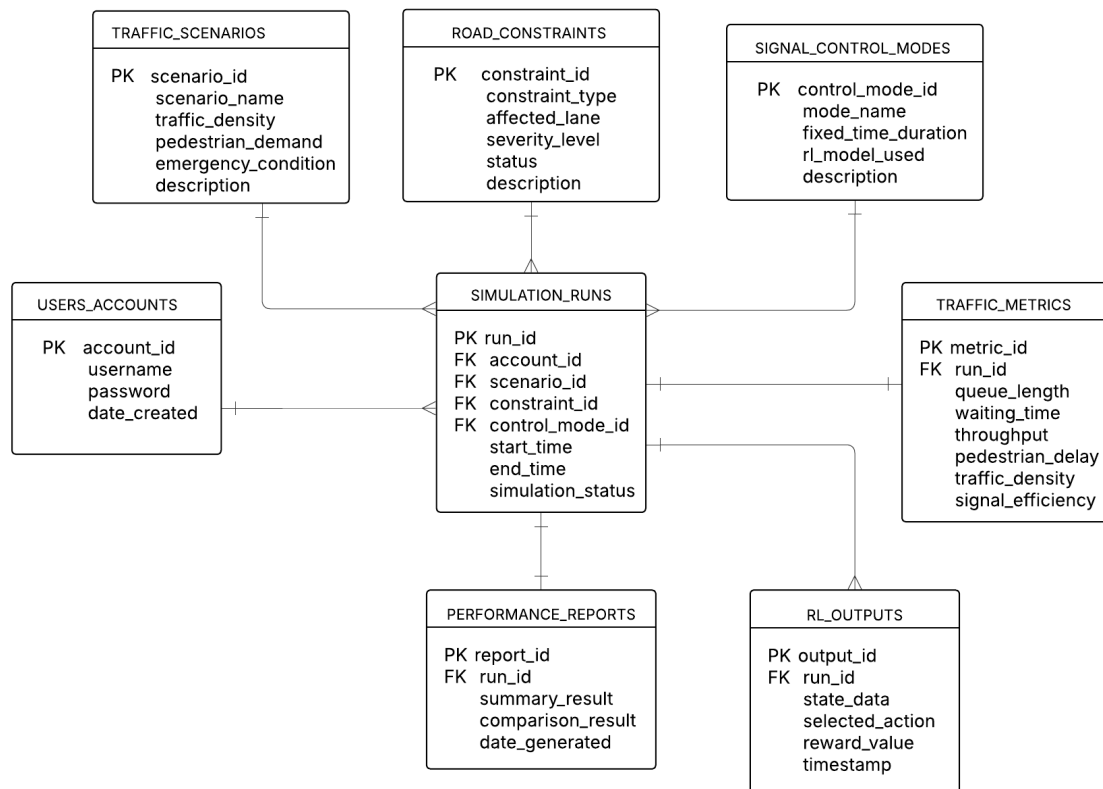


Figure 16. Entity Relationship Diagram

Data Dictionary

The data dictionary of the SMARTFLOW Traffic System provides a detailed description of the data elements used in the system's SQLite local database. It serves as a reference for understanding how data is organized, stored, and managed to support system functions such as user authentication, traffic scenario configuration, road constraint management, simulation execution, reinforcement learning output tracking, and performance reporting.

The system is composed of several tables, including Users, Traffic Scenarios, Road Constraints, Signal Control Modes, Simulation Runs, Traffic Metrics, RL Outputs, and Performance Reports. Each table contains fields with defined data types, constraints, and descriptions to ensure consistent data storage and accurate retrieval of simulation-related information.

Table 2. User

Field Name	Data Type	Length	Constraints	Description
accuont_id	INTEGER	-	PRIMARY KEY	Unique identifier for each authorized user
username	TEXT	50	UNIQUE, NOT NULL	Username used for local system login
password	TEXT	255	NOT NULL	Password used for authentication
date_created	DATETIME	-	NOT NULL	Date and time the user record was created

Table 3. Traffic Scenarios

Field Name	Data Type	Length	Constraints	Description
scenario_id	INTEGER	-	PRIMARY KEY	Unique identifier for each traffic scenario
scenario_name	TEXT	100	NOT NULL	Name of the traffic scenario
traffic_density	TEXT	30	NOT NULL	Traffic density classification such as normal or heavy
pedestrian_demand	TEXT	30	NOT NULL	Level of pedestrian activity in the simulation
emergency_condition	TEXT	30	NULL	Indicates presence of emergency vehicle conditions
description	TEXT	255	NULL	Additional description of the scenario

Table 4. Road Constraints

Field Name	Data Type	Length	Constraints	Description
constraint_id	INTEGER	-	PRIMARY KEY	Unique identifier for each road constraint

constraint_type	TEXT	50	NOT NULL	Type of road constraint such as lane closure or accident
affected_lane	TEXT	50	NOT NULL	Lane or road section affected by the constraint
severity_level	TEXT	20	NOT NULL	Severity classification of the constraint
status	TEXT	20	NOT NULL	Current status of the road constraint
description	TEXT	255	NULL	Additional details regarding the constraint

Table 5. Signal Control Modes

Field Name	Data Type	Length	Constraints	Description
control_mode_id	INTEGER	-	PRIMARY KEY	Unique identifier for each control mode
mode_name	TEXT	50	NOT NULL	Name of the signal control mode
fixed_time_duration	INTEGER	-	NULL	Preset signal duration for fixed-time control
rl_model_used	TEXT	100	NULL	Name or identifier of the RL model used
description	TEXT	255	NULL	Additional information about the control mode

Table 6. Simulation Runs

Field Name	Data Type	Length	Constraints	Description
run_id	INTEGER	-	PRIMARY KEY	Unique identifier for each simulation run
account_id	INTEGER	-	FOREIGN KEY	Reference to the authorized user
scenario_id	INTEGER	-	FOREIGN KEY	Reference to the selected traffic scenario
constraint_id	INTEGER	-	FOREIGN KEY	Reference to the applied road constraint
control_mode_id	INTEGER	-	FOREIGN KEY	Reference to the selected signal control mode
start_time	DATETIME	-	NOT NULL	Simulation start timestamp
end_time	DATETIME	-	NULL	Simulation end timestamp
simulation_status	TEXT	30	NOT NULL	Current status of the simulation

Table 7. Traffic Metrics

Field Name	Data Type	Length	Constraints	Description
metric_id	INTEGER	-	PRIMARY KEY	Unique identifier for traffic metric data

run_id	INTEGER	-	FOREIGN KEY	Reference to the simulation run
queue_length	REAL	-	NOT NULL	Measured vehicle queue length
waiting_time	REAL	-	NOT NULL	Average waiting time during simulation
throughput	REAL	-	NOT NULL	Number of vehicles passing through the intersection
pedestrian_delay	REAL	-	NULL	Delay experienced by pedestrians
traffic_density	REAL	-	NULL	Measured traffic density
signal_efficiency	REAL	-	NULL	Efficiency score of signal control

Table 8. RL Outputs

Field Name	Data Type	Length	Constraints	Description
output_id	INTEGER	-	PRIMARY KEY	Unique identifier for RL output data
run_id	INTEGER	-	FOREIGN KEY	Reference to the simulation run

state_data	TEXT	500	NOT NULL	Encoded traffic state information
selected_action	TEXT	100	NOT NULL	Traffic signal action selected by the RL model
reward_value	REAL	-	NOT NULL	Reward value generated during RL processing
timestamp	DATETIME	-	NOT NULL	Date and time the RL action was generated

Table 9. Performance Reports

Field Name	Data Type	Length	Constraints	Description
report_id	INTEGER	-	PRIMARY KEY	Unique identifier for each performance report
run_id	INTEGER	-	FOREIGN KEY	Reference to the simulation run
summary_result	TEXT	1000	NOT NULL	Summary of simulation performance results
comparison_result	TEXT	1000	NULL	Comparison between fixed-time and RL-based dual control
generated_by	TEXT	100	NULL	User or process that generated the report

date_generated	DATETIME	-	NOT NULL	Date and time the report was generated
----------------	----------	---	----------	--

Technologies, Concepts, and Theories

This section presents the key models, algorithms, concepts, and technologies utilized in the development of the SMARTFLOW Traffic System. It explains how traffic simulation data is generated, processed, analyzed, and used for reinforcement learning–based traffic signal optimization. The discussion follows the major stages of the system workflow, including data collection, data pre-processing, feature extraction, reinforcement learning–based decision-making, and system implementation technologies.

Data Collection

Data collection in SMARTFLOW is performed within the SUMO simulation environment. Instead of using live CCTV feeds, IoT sensors, or GPS devices, the system gathers traffic-related data generated during simulation runs. These data include vehicle count, pedestrian count, queue length, waiting time, traffic density, throughput, pedestrian delay, emergency vehicle presence, and traffic signal states. Through TraCI, Python retrieves these simulation values in real time and forwards them to the control and decision-making components of the system.

Data Pre-Processing

Before the collected simulation data is used by the reinforcement learning controller, it undergoes pre-processing to ensure that it is organized, consistent, and suitable for analysis. This includes formatting traffic state values, filtering unnecessary simulation outputs, normalizing numerical values such as queue length and waiting time, and converting raw traffic data into a structured observation state. Pre-processing helps reduce errors and ensures that the

reinforcement learning model receives clean and meaningful input during training and evaluation.

Feature Extraction

Feature extraction involves identifying the most relevant traffic attributes that influence signal control decisions. In SMARTFLOW, the extracted features may include queue length per lane, average waiting time, vehicle density, pedestrian demand, emergency vehicle presence, current signal phase, and road constraint status. These features represent the current traffic condition of the simulated intersection and serve as the input state for the reinforcement learning model.

Reinforcement Learning–Based Decision-Making

The SMARTFLOW Traffic System uses Reinforcement Learning as the decision-making approach for adaptive traffic signal control. In the system, the traffic signal controller acts as the learning agent, while the SUMO simulation environment serves as the traffic environment where traffic conditions are observed and signal actions are applied through TraCI.

The agent observes the current traffic state of the intersection, including queue length, vehicle count, average waiting time, traffic density, pedestrian demand, emergency vehicle presence, road constraint status, and the current traffic signal phase. Based on this state, the agent selects a signal action, such as changing to the next valid traffic phase or maintaining or extending the current phase depending on the traffic condition.

Instead of using a fixed signal timing, the system allows the agent to learn from different traffic situations and evaluate whether its chosen signal action improves traffic flow. The learning process considers important traffic performance factors such as vehicle waiting time, queue length, pedestrian delay, traffic throughput, and emergency vehicle priority. Actions that help reduce congestion, minimize delay, improve vehicle movement, and prioritize emergency

vehicles are treated as more favorable, while actions that increase waiting time, queue buildup, or pedestrian delay are considered less effective.

The training process is performed using simulation-generated data from repeated SUMO episodes rather than live CCTV, GPS, or physical sensor data. Each episode represents a complete simulation run under a selected traffic scenario, such as normal traffic, heavy congestion, pedestrian-intensive conditions, emergency vehicle conditions, lane closures, road construction, or accidents.

During training, SUMO generates the traffic conditions, TraCI transfers the observed traffic state to the reinforcement learning component, the agent selects a signal action, and the system evaluates the result based on the traffic performance outcome. Through repeated simulation experiences, the agent improves its decision-making and learns how to select signal actions that produce better traffic performance.

In this study, the reinforcement learning–based signal control approach will be tested and compared against a fixed-time traffic signal control baseline to determine whether the adaptive system can improve waiting time, queue length, throughput, pedestrian delay, and emergency vehicle response.

Technologies Used in the System

Simulation of Urban Mobility (SUMO)

The system primarily utilizes Simulation of Urban Mobility (SUMO), an open-source traffic simulation platform used to model and simulate road networks, vehicle movement, pedestrian behavior, and traffic signal operations. SUMO serves as the main simulation environment of the SMARTFLOW system, allowing the proponents to create realistic traffic conditions, road constraint scenarios, and traffic flow behavior within the selected intersection in Tagum City.

Through SUMO, the system can simulate different traffic densities, pedestrian activity, emergency vehicle scenarios, lane closures, road construction, accidents.

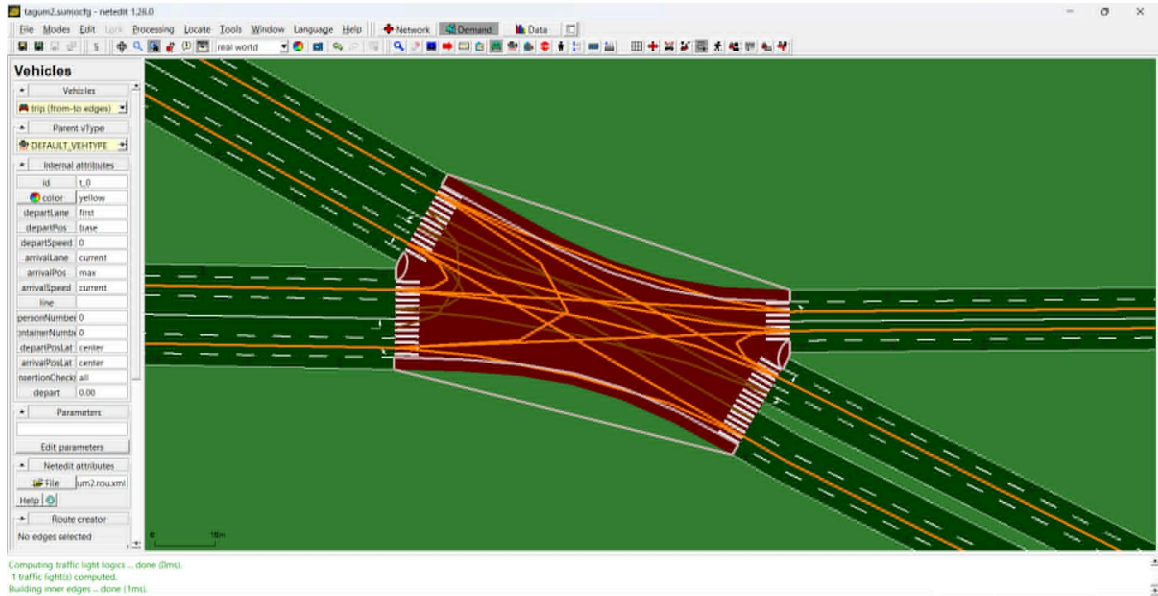


Figure 17. Simulation of Urban Mobility (SUMO)

Reinforcement Learning (RL)

To enable adaptive traffic signal optimization, the system integrates Reinforcement Learning (RL) techniques. Reinforcement Learning allows the system to learn optimal traffic signal control strategies by continuously interacting with the simulation environment. The RL model observes traffic conditions, performs signal control actions, receives reward feedback, and improves its decision-making performance over repeated simulation episodes. This approach enables the SMARTFLOW system to dynamically adapt traffic signals based on changing traffic flow, pedestrian demand, and road constraint conditions, improving overall traffic efficiency compared to traditional fixed-time traffic signal control.

```

1 from __future__ import annotations
2
3 import json
4 import statistics
5 from dataclasses import asdict, dataclass
6 from datetime import UTC, datetime
7 from pathlib import Path
8 from typing import Callable
9
10 try:
11     from stable_baselines3 import DQN
12     from stable_baselines3.common.callbacks import BaseCallback
13 except ImportError: # pragma: no cover - exercised when the local venv lacks the RL stack
14     DQN = None
15     BaseCallback = None
16
17
18 OBSERVATION_VERSION = "obs_v1"
19 REWARD_VERSION = "reward_v1"
20 ACTION_SPACE_VERSION = "action_v1"
21
22
23 @dataclass(frozen=True)
24 class DQNTTrainingConfig:
25     total_timesteps: int
26     learning_rate: float = 1e-4
27     buffer_size: int = 50_000
28     learning_starts: int = 100
29     batch_size: int = 32
30     gamma: float = 0.99
31     train_freq: int = 1
32     gradient_steps: int = 1
33     target_update_interval: int = 250
34     exploration_fraction: float = 0.2
35     exploration_initial_eps: float = 1.0
36     exploration_final_eps: float = 0.05
37     seed: int = 0

```

Figure 18. Reinforcement Learning (RL)

Python and TraCI

Python will serve as the main programming language for connecting the simulation, reinforcement learning controller, dashboard logic, and database operations. Its module-based structure supports organized development and integration of system components. TraCI, or Traffic Control Interface, will be used because it provides programmatic access to a running SUMO simulation and allows Python to retrieve simulation values and send traffic signal control commands in real time. Together, Python and TraCI are relevant to SMARTFLOW because they form the communication layer between the SUMO environment and the RL-based decision-making component.

```

111 class SumoSimulationEngine:
234     def configure(self, **kwargs):
245         if kwargs:
247             def configure_from_scenario(self, scenario: dict):
248                 if scenario.get('intersection_id'):
249                     self.set_intersection(scenario['intersection_id'])
250                 if scenario.get('name'):
251                     self.current_scenario_name = scenario['name']
252                 self.configure(**{
253                     key: scenario[key]
254                     for key in (
255                         "traffic_density",
256                         "pedestrian_density",
257                         "emergency_mode",
258                         "road_constraint",
259                         "lane_closure",
260                         "construction",
261                         "accident",
262                         "flooding",
263                         "temp_blockage",
264                     )
265                     if key in scenario and scenario[key] is not None
266                 })
267             def configure_rl_control(
268                 self,
269                 ",
270                 decision_interval: float = 5.0,
271                 minimum_green_hold: float = 10.0,
272                 controller_label: str = "RL Controller",
273                 controller_provenance: str = "rl",
274             ):
275                 self.rl_control_enabled = True
276                 self.rl_decision_interval = max(float(decision_interval or 5.0), SUMO_STEP_LENGTH)
277                 self.rl_minimum_green_hold = max(float(minimum_green_hold or 10.0), SUMO_STEP_LENGTH)
278                 self.rl_pending_action = None

```

Figure 19. Python and TraCI

PyTorch

PyTorch will be used to implement and train the Deep Q-Network (DQN) model for the reinforcement learning component of the SMARTFLOW Traffic System. It provides optimized tensor operations and deep learning support that are suitable for training neural network-based models on CPUs and GPUs. In SMARTFLOW, PyTorch is relevant because it will be used to build the DQN model that estimates the best traffic signal action based on the current simulation state, such as queue length, waiting time, traffic density, pedestrian demand, emergency vehicle presence, and road constraint conditions. It will also support model training by processing simulation experiences generated from SUMO episodes, computing training loss, and updating model parameters to improve adaptive traffic signal decision-making.

```

C:\Users\jhepo\Desktop> trapik2 > simulation > ♦ n.policy.runtime.py > ❷ extract_dql_q_values
89 def masked_argmax(values: Iterable[float], valid_action_mask: tuple[int, ...] | None = None) -> int:
96     for index, value in enumerate(q_values):
97         if valid_action_mask is not None and (index >= len(valid_action_mask) or not valid_action_mask[index]):
98             continue
99         if best_value is None or value > best_value:
100             best_action = index
101             best_value = value
102     if best_action is None:
103         return first valid action(valid_action_mask)
104     return int(best_action)
105
106
107 def _extract_dql_q_values(model, observation) -> tuple[float, ...]:
108     if hasattr(model, "predict_q_values"):
109         return tuple(float(value) for value in model.predict_q_values(observation))
110
111     try:
112         import numpy as np
113         import torch
114     except ImportError:
115         return ()
116
117     if observation is None or not hasattr(model, "q_net"):
118         return ()
119
120     try:
121         observation_array = np.asarray(observation, dtype=np.float32)
122         if observation_array.ndim == 1:
123             observation_array = observation_array.reshape(1, 1)
124         observation_tensor = torch.as_tensor(observation_array, device=model.device)
125         with torch.no_grad():
126             q_values = model.q_net(observation_tensor)
127         return tuple(float(value) for value in q_values.detach().cpu().numpy()[0])
128     except Exception:
129         return ()
130

```

Figure 20. PyTorch

Database Management System

In terms of data management, the SMARTFLOW Traffic System utilizes SQLite as its Database Management System (DBMS) to store and organize user credentials, traffic scenarios, road constraints, simulation results, reinforcement learning outputs, and performance reports. This allows efficient data retrieval, simulation monitoring, and performance evaluation within the system. Additionally, the database supports local storage without requiring internet connectivity or a centralized server, making the system lightweight, reliable, and suitable for simulation-based traffic analysis and adaptive traffic signal optimization.



```
1 """
2 SMARTFLOW - SQLite Database Layer
3 All schema definitions, CRUD operations, and seed data.
4 """
5
6 import sqlite3
7 import os
8 import json
9 import secrets
10 from datetime import UTC, datetime, timedelta
11 from contextlib import contextmanager
12 from pathlib import Path
13
14 import config
15
16 SCENARIO_JSON_FIELDS = (
17     "lane_closure_config",
18     "construction_config",
19     "accident_config",
20     "flooding_config",
21 )
22 AUTH_SESSION_VERSION_SETTING = "auth_session_version"
23
24
25 def get_connection():
26     os.makedirs(os.path.dirname(config.DB_PATH), exist_ok=True)
27     conn = sqlite3.connect(config.DB_PATH)
28     conn.row_factory = sqlite3.Row
29     conn.execute("PRAGMA journal_mode=WAL")
30     conn.execute("PRAGMA foreign_keys=ON")
31     conn.execute("PRAGMA busy_timeout=5000")
32     return conn
33
34
35 def _connect_sqlite_file(path: str | Path) -> sqlite3.Connection:
36     conn = sqlite3.connect(str(path))
37     conn.row_factory = sqlite3.Row
```

Figure 21. Database Management System (DBMS)

SYSTEM TESTING AND IMPLEMENTATION

The System Testing and Implementation phase focuses on ensuring that the SMARTFLOW Traffic System operates properly and satisfies the specified system requirements. During this phase, comprehensive testing is conducted to validate the integration of the major system components, including the SUMO simulation environment, Python-TraCI communication layer, reinforcement learning-based traffic signal controller, SQLite database, and dashboard interface. The testing process evaluates the performance of the adaptive traffic signal optimization system using metrics such as average waiting time, queue length, throughput, traffic density, pedestrian delay, and signal efficiency to ensure accurate and responsive traffic management under varying traffic and road constraint conditions.

The SMARTFLOW system will undergo unit testing, integration testing, and scenario-based testing to verify the functionality of individual components and the overall system. Test cases are designed to simulate both normal and disruptive traffic scenarios, including heavy traffic congestion,

pedestrian-intensive conditions, lane closures, road construction, accidents, and emergency vehicle situations. These tests verify whether the system can correctly process simulation data, apply adaptive traffic signal decisions, and maintain stable simulation performance throughout execution.

In addition to functional testing, the Reinforcement Learning (RL)–based traffic signal controller will be evaluated using RL-specific performance metrics, including cumulative reward, convergence rate, policy stability, average waiting time, queue length, and traffic throughput. These metrics will be used to assess the learning efficiency, adaptability, and effectiveness of the RL agent in optimizing traffic signal decisions under different traffic and road constraint scenarios. The results will also be compared with fixed-time traffic signal control to determine the overall performance improvement achieved by the RL model.

A summary of the testing results should be presented in tabular form, highlighting the status and performance of each major component, such as traffic simulation, reinforcement learning processing, signal control execution, database storage, dashboard monitoring, and performance reporting. This allows the proponents to identify issues and confirm that the functionalities of the SMARTFLOW system meet the expected operational standards.

To support the validation process, a System Test Plan should be prepared outlining the objectives, testing scope, simulation environment, test cases, procedures, and acceptance criteria. This ensures that the testing activities remain organized, systematic, and aligned with the functional and non-functional requirements of the study.

Following successful testing, the implementation phase involves deploying the SMARTFLOW system within a localized desktop simulation environment. This includes configuring the SUMO simulation platform, integrating Python and TraCI communication, implementing the reinforcement learning model, setting up the SQLite database, and deploying the local dashboard interface. The deployed system will be used by researchers and authorized traffic management personnel

to evaluate traffic scenarios, monitor simulation performance, and assess the effectiveness of the Reinforcement Learning–based traffic signal controller before any real-world implementation. After deployment, final validation is conducted to confirm that the system performs correctly within the intended operating environment. Users may also be oriented on the proper use of the simulation dashboard and traffic analysis features.

Finally, continuous monitoring and evaluation are conducted to assess system performance, identify simulation issues, and improve adaptive traffic signal optimization strategies. This phase ensures that the SMARTFLOW system remains functional, reliable, and adaptable for future traffic simulation studies and intelligent transportation system research.

SYSTEM MAINTENANCE

The System Maintenance phase focuses on ensuring the continued reliability, functionality, and improvement of the SMARTFLOW Traffic System after implementation. This phase involves monitoring system performance, identifying and correcting system issues, and applying necessary updates to improve simulation accuracy, reinforcement learning performance, and overall usability.

Maintenance activities may include regular checking of the SUMO simulation environment, updating Python and reinforcement learning components, refining traffic scenario and road constraint configurations, improving dashboard features, and maintaining the SQLite database. The proponents may also adjust simulation parameters, optimize reinforcement learning behavior, and correct detected errors based on testing results and user feedback.

Additionally, continued evaluation of traffic simulation performance and adaptive signal control behavior may be conducted to support future

enhancements and expanded traffic analysis scenarios. This phase ensures that SMARTFLOW remains stable, maintainable, and suitable for future development in intelligent traffic management and simulation-based decision-support research.

System Security Plan

To ensure the protection of system data and simulation resources, a System Security Plan will be implemented for the SMARTFLOW Traffic System. The system will enforce user authentication and role-based access control to prevent unauthorized access to the simulation dashboard, database, and system configurations. Since the system operates within a localized desktop environment, locally stored simulation data, reinforcement learning outputs, and configuration files will be protected through secure access management and regular database backup procedures.

The system will also apply secure communication and processing practices between the SUMO simulation environment, Python-TraCI communication layer, reinforcement learning components, and SQLite database. Input validation, error handling, and secure coding practices will be implemented to reduce vulnerabilities and ensure accurate traffic simulation processing. In addition, audit logs and monitoring functions may be used to track user activities, simulation execution, and system events for security and troubleshooting purposes.

The System Security Plan shown in Table 4 follows ISO/IEC 27001–aligned security practices to support the confidentiality, integrity, and availability of the SMARTFLOW Traffic System. These security controls help ensure secure system operation, reliable simulation processing, and protected management of traffic simulation data and reinforcement learning outputs.

Table 10. System Security Plan

Security Domain (ISO 27001)	Control Area	Description / Implementation in the System
A.5 Information Security Policies	Security Policy	The system will implement security policies for user access, simulation control, and local data protection.
A.6 Organization of Information Security	Roles and Responsibilities	Roles such as Administrator, Researcher, and Authorized Operator will have defined system privileges.
A.9 Access Control	User Access Management	The system will enforce role-based access control (RBAC) through secured login credentials.
A.9 Access Control	Authentication Mechanism	Username and password authentication will be implemented to restrict unauthorized access.
A.10 Cryptography	Data Protection	Local simulation data and exported files will be protected through secure storage and backup mechanisms.

A.12 Operations Security	Data Processing Integrity	Input validation and error handling will ensure accurate traffic simulation processing and stable system operation.
A.12 Operations Security	Logging and Monitoring	The system will maintain logs of simulation activities, user actions, and system events for monitoring and troubleshooting.
A.13 Communications Security	Secure Communication	Communication between SUMO, Python, TraCI, and reinforcement learning components will follow secure processing practices.
A.11 Physical and Environmental Security	Device Protection	Desktop devices and local storage used for simulation execution will be secured from unauthorized physical access.
A.17 Information Security Aspects of Business Continuity	Data Backup and Recovery	Regular backup of SQLite databases and simulation reports will be performed to ensure data recovery.

A.16 Information Security Incident Management	Incident Response	Procedures will be established for detecting, reporting, and resolving system errors or security-related issues.
A.14 System Acquisition, Development, and Maintenance	Secure Development	Secure coding practices, debugging, and system updates will be applied throughout development and maintenance.

System Maintenance Plan

In addition, a System Maintenance Plan will be established to support the continued operation, improvement, and sustainability of the SMARTFLOW Traffic System after implementation. This plan includes scheduled maintenance activities such as simulation optimization, reinforcement learning model refinement, SQLite database maintenance, dashboard improvement, and system performance monitoring. Procedures for troubleshooting, debugging, and correcting system errors will also be implemented to maintain stable simulation performance.

Furthermore, system documentation will be continuously updated to record configuration changes, reinforcement learning adjustments, simulation updates, and maintenance activities. This ensures easier future enhancement, maintainability, and expansion of the SMARTFLOW system for additional traffic scenarios and intelligent transportation studies.

The System Maintenance Plan shown in Table 11 aligns maintenance activities with ISO/IEC 25010 software quality attributes to ensure that the

SMARTFLOW Traffic System remains reliable, efficient, maintainable, secure, and adaptable over time.

Table 11. System Maintenance Plan

ISO 25010 Attribute	Maintenance Activity	Description	Frequenc y
Reliability	System Monitoring	Monitoring simulation execution, system stability, and dashboard operation	Continuou s
Performance Efficiency	Performance Optimization	Improving simulation processing speed and reinforcement learning performance	Regularly
Usability	Dashboard Improvement	Updating dashboard layout and controls based on user feedback	As Needed
Security	Security Updates	Applying system updates, access control improvements, and database backup procedures	Periodicall y

Maintainability	Debugging and Code Refactoring	Correcting bugs and improving system structure for easier maintenance	Regularly
Compatibility	System Updates	Ensuring compatibility with SUMO, Python, PyTorch, and desktop environments	As Needed
Reliability	Simulation Calibration	Adjusting traffic simulation parameters and reinforcement learning configurations	Periodically
Performance Efficiency	Database Optimization	Managing SQLite database cleanup, indexing, and report storage optimization	Monthly

REFERENCES:

- [1] Injac, Z., Arsić, S., Drašković, D., & Arsić, M. (2025). Urban traffic management using artificial intelligence: A sustainable approach to enhancing urban mobility. *Journal of Traffic and Transport Theory and Practice (JTTP)*, 10(1), 30-35.
- [2] Akinade, A.O., Adepoju, P.A., & Ige, A.B. Artificial Intelligence in Traffic Management: A Review of Smart Solutions and Urban Impact.
- [3] Rathore, S. P. S., Farhaoui, Y., Aniebonam, E. E., Nagpal, T., & Kaushik, P. (2025, March). AI-driven traffic congestion management: a predictive analytics approach for smart cities. In 2025 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI) (Vol. 3, pp. 1-6). IEEE.
- [4] S. Rudhra, S. Vinod, P. Arthi and L. Kurinjimalar, "Density Based Traffic Signal System," 2024 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS), Chennai, India, 2024, pp. 1-3, doi: 10.1109/ICPECTS62210.2024.10780421.

- [5] Gheorghe, C., & Şoica, A. (2025). Revolutionizing Urban Mobility: A Systematic Review of AI, IoT, and Predictive Analytics in Adaptive Traffic Control Systems for Road Networks. *Electronics*.
- [6] Kodama, N., Harada, T., & Miyazaki, K. (2022). Traffic signal control system using deep reinforcement learning with emphasis on reinforcing successful experiences. *IEEE Access*, 10, 128943-128950.
- [7] Huang, S. Y., Chang, H. C., Chen, Y. C., Wei, T. H., Yeh, I. H., Kuan, S. Y., ... & Wu, I. (2026). A Robust and Efficient Multi-Agent Reinforcement Learning Framework for Traffic Signal Control. *arXiv preprint arXiv:2603.12096*.
- [8] Almomany, A., Eedi, E., & Sutcu, M. (2025). Real-time traffic signal optimization for urban mobility: a reinforcement learning-enhanced framework with application to Kuwait City. *Frontiers in Robotics and AI*, 12, 1669952.
- [9] United Nations, "Goal 9: Build resilient infrastructure, promote sustainable industrialization and foster innovation," *Sustainable Development Goals*, 2023.
- [10] United Nations, "Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable," *Sustainable Development Goals*, 2023.
- [11] M. Muralidharan and R. Pedarsani, "Analysis of fixed-time control in signalized intersection networks," *IEEE Transactions on Control of Network Systems*, vol. 2, no. 1, pp. 22-31, 2014.
- [12] Sylva, V., & Lala, A. (2026). Microsimulation-Based Evaluation of Fixed-Time and Vehicle-Actuated Traffic Signal Control Strategies Using PTV Vissim in an Urban Corridor. *Informatica*, 50(5).
- [13] C. Aboy, M. Chua, S. Gochuico, J. Nuesca, and M. Suwanpimol, "PC-Based Adaptive Road Traffic Control System," Undergraduate thesis, Dept. Electron. Commun. Eng., De La Salle Univ., Manila, Philippines, 1994.
- [14] Geotab, "Video Intelligence and AI-driven enforcement in the Philippines," *PR Newswire Asia*, May 2026.
- [15] Grand View Research, "Philippines Intelligent Transportation System Market Size & Outlook,"

- [16] DOST-PCIEERD, "An Analysis of Smart City Development Frameworks in the Philippines," UP CIDS Policy Brief
- [17] Dubey, A., Lakhani, M., Dave, S., & Patoliya, J. J. (2017, December). Internet of Things based adaptive traffic management system as a part of Intelligent Transportation System (ITS). In 2017 international conference on soft computing and its engineering applications (icSoftComp) (pp. 1-6). IEEE.
- [18] Zemmouchi-Ghomari, L. (2025). Artificial intelligence in intelligent transportation systems. *Journal of Intelligent Manufacturing and Special Equipment*, 6(1), 26-42.
- [19] Damadam, S., Zourbakhsh, M., Javidan, R., & Faroughi, A. (2022). An Intelligent IoT Based Traffic Light Management System: Deep Reinforcement Learning. *Smart Cities*, 5 (4), 1293-1311.
- [20] Kodama, N., Harada, T., & Miyazaki, K. (2022). Traffic signal control system using deep reinforcement learning with emphasis on reinforcing successful experiences. *IEEE Access*, 10, 128943-128950.
- [21] H. Zhang, "Multi-Agent Reinforcement Learning Framework for Traffic Signal Control," arXiv preprint arXiv:2603.12096, 2026.
- [22] T. Miller, "Quantifying the Impact of RL on Urban Congestion," *Journal of Urban Systems*, vol. 15, no. 4, pp. 201-215, 2025.
- [23] A. Al-Wajih, "AI in ITS: Real-Time Decision-Making Architectures," *IEEE Access*, vol. 13, pp. 14502-14518, 2025.
- [24] M. J. Williams, "Agent-Based Modeling of Urban Mobility Landscapes," *IEEE Transactions on Smart Cities*, vol. 9, no. 2, pp. 112-125, 2024.
- [25] D. Krajzewicz, "SUMO - Simulation of Urban MObility," *IEEE Intelligent Transportation Systems Conference*, 2012.
- [26] NEDA, "Philippine Development Plan 2023-2028: Digital Transformation for Connectivity," 2023.
- [27] Kodama, N., Harada, T., & Miyazaki, K. (2022). Traffic signal control system using deep reinforcement learning with emphasis on reinforcing successful experiences. *IEEE Access*, 10, 128943-128950.
- [28] S. Suo, S. Regalado, S. Casas, and R. Urtasun, "TrafficSim: Learning to Simulate Realistic Multi-Agent Behaviors," in *Proceedings of the*

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021, pp. 10400-10409.

- [29] K. AlMulla, S. Ashkanani, F. Hassan, H. AlMesbahi, M. Alazmi, and M. Nadeem, "IoT-Based Adaptive Traffic Signal Controller to Optimize the Flow of Traffic and Reduce Congestion," in *Proceedings of the 2024 1st Mediterranean Smart Cities Conference (MSCC)*, 2024